



Comprehensive literature review on children automatic speech recognition system, acoustic linguistic mismatch approaches and challenges

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Abstract

Automatic Speech Recognition (ASR) system for children is as important as for adults since children are more dependent on these systems nowadays, such as computer games, reading tutors, foreign language learning tools, etc. Consequently, this article aims to present several important aspects related to children's speech recognition systems, in which a comprehensive review is presented. Acoustic and linguistic challenges of children's speech are presented thoroughly to understand the basic anatomy of children's articulation organs. A variety of challenges exist for the development of children's ASR, such as the collection of children's speech data is a very complex task; the available child corpora are not publicly accessible, children's speakers differ greatly due to linguistic and acoustic variations, and ASRs developed for one age group are not suitable for another age group. All these challenges are systematically described in this article. Various data augmentation methods are also explored here, along with different approaches to develop ASR in children's speech. It has been observed that the inaccessibility of child corpora publicly is a significant barrier to children's ASR. Apart from the challenges mentioned earlier related to children's ASR, an attempt has been made to thoroughly review the children's ASR in the case of Punjabi language, as this language is ranked 10th most spoken globally and is still considered a low-resource language. Further, various approaches for the development of children's ASR such as traditional, hybrid and end-to-end (E2E) networks are also reported. In addition, an analytical summary and discussion are included.

Keywords Automatic Speech Recognition · Applications of Child ASRs · Data Augmentation · Acoustic and Linguistic Variations · Mismatch ASR · Low Resource Language

1 Introduction

People communicate primarily through speech, which is a natural and primary method of communication. Being a natural source of communication, the speech also provides efficient communication between humans and machines [1]. It is the only source that acts as

an interface to make the best use of smart devices using various applications. Moreover, it can also be used for accessing any information, wherein the user interacts with the system using a voice command, and the system recognizes the command and reacts accordingly. For any ASR, humans must store their speech patterns and vocabulary in the system in order to train the system. It is essential to combine understanding from various fields, such as signal processing, audio, computer science, linguistics, psychology, communication theory, physiology, and statistics, to conduct research in ASR [2–4]. The embedded ASR must act as an interface that can make the use of any smart device quite efficient. Speech Interpretation and Recognition Interface (SIRI) by Apple, Google assistance from Google, Cortana by Microsoft, and Alexa by Amazon are some well-known examples of ASR. Since adults are expected to be the primary users of the developed technology, many speech interfaces are developed specifically for adults. Commercial ASR systems for adults generally work well [5–7]. It was possible to use the voices of both males and females in these systems. However, many growing children are exposed to advanced technology at a younger age today. It has been observed that about 78% (This number would have increased even more at the time of Covid-19) of children (aged 6 to 9 years) use the internet for their homework and playing games [8]. It is becoming increasingly important to develop children's speech recognition systems because of their applications in education & assessment, entertainment, and medical fields. As a result, child ASR has received more attention in recent years. Despite numerous ASR applications, researchers have yet to achieve 100% accuracy in the system. A number of factors can affect the spectral and temporal characteristics of speech waves which can lead to variabilities in speech. Various classifications of variability are available in the literature [9–13], which includes *extrinsic* and *intrinsic* variabilities. Extrinsic variability is associated with external factors/environment, which includes a signal-to-noise ratio, channels, and microphones. However, these variabilities cannot be eliminated completely but are still controllable, whereas intrinsic variability is associated only with the speaker e.g., gender, age, the anatomy of the vocal tract, social & regional origin, health, and emotional state. Intrinsic variabilities, including speaker-specific properties and linguistic variabilities of the speaker, affect the speech signals at large extent and hence ASR systems. These are very hard to control comparatively [9, 10].

The contribution of the article is as mentioned below:

1. The article highlights various challenges in developing children's ASR, which include the inaccessibility of child corpora publicly, which is a major barrier to children's ASR. Variability of children speakers, non-availability of child corpus, and difficulties in child speech data collection are some of the significant challenges for fewer children-ASR.
2. The article presents various acoustic and linguistic challenges of children's speech to understand the basic anatomy of children's articulation organs, which must be considered while designing a child ASR system.
3. Acoustic and Linguistic mismatch approaches for child ASR have been discussed. Further, detailed elaboration of acoustic mismatch based on feature, signal, and model domains has been presented. A comparison of various existing approaches of acoustic mismatch has been provided based on ASR parameters.
4. Data augmentation approaches for Child ASR system and language-based review of child ASR system wherein literature review for the low resource Punjabi language has been presented. The article also provides an extensive analytical summarization and discussion on the various approaches used for child ASR.

The paper is organized as follows: Section 2 discusses the various applications of child ASR. Section 3 presents different challenges in child ASR. Section 4 elaborates on various characteristics of children's speech. Details of different approaches used for child ASR are discussed in Section 5. Section 6 summarizes the data augmentation techniques used for speech enhancement in children ASR. Section 7 elaborates on low resource languages and review of children's ASR in the Punjabi language. Section 8 presents analytical summary and discussion. Section 9 finally concludes the paper.

2 Applications of child ASR

Developing speech recognition systems for children is becoming increasingly important because of two main reasons: *i)* due to their applications in education and assessment, foreign language learning tools, entertainment, and medical & Clinical. *ii)* the use of a speech interface is beneficial for children who cannot operate traditional input devices such as a keyboard, mouse, and pointer [5]. However, it takes more time and effort to design an ASR system for children's speech than for adults' speech [14].

Numerous applications of ASR system have been developed in *education and assessment* over the past several years. Previously, a computerized speech training tool called Speech Training Aid Research (STAR) was used to teach children how to speak. The tool was marketed by Hereford and Worcester County Council (HWCC). A child's pronunciation of a word was evaluated using STAR either as poor or good according to whether it was spoken with a limited vocabulary at the beginning. Other learning project "Talking Books" was developed by Sherston software. They successfully integrate speech technology into their products. These are used to read a story aloud to a youngster and can listen to the child's reading and correct the child's incorrectly pronounced words [15–17]. For improving children's reading skills for several reasons, a system based on ASR technology may be effective [18]. In *language learning*, ASRs also encourage children to learn foreign languages. When a child learns to speak or read, the learning process can be enhanced by significant reading practice with individual support from a tutor. But in reality, giving full attention to a single child is not possible for a single tutor/ teacher. Therefore, additional use of a fully automated reading tutor would be beneficial for the child and of course for the teacher. "Say it again, kid!" (SIAC) is a well-known application for learning a foreign language. With the help of the game, children will become familiar with foreign vocabulary and pronunciation and will learn how to listen and speak as well.

With the Automatic Reading Tutor (ART), children are provided with appropriate feedback on where the words are in relation to the story text in their oral reading. It has been shown that even providing simple tracking feedback to a child can help keep them engaged in oral reading. ART systems can also detect common reading errors for beginning readers, such as mispronunciations, repetitions, deletions, filler pauses, etc. As a result of correcting those miscues, the system may be able to provide the child with the appropriate help information. As part of this, pre-recorded or Text-To-Speech (TTS) -based pronunciations are provided to correct the mispronunciation, along with suggestions regarding specific practices. Additionally, the ART system can measure a child's level of fluency and provide test statistics similar to regular language tests. Compared to human tutors, ART systems maintain children's interest in reading and learning for longer periods and improve reading proficiency than traditional classroom reading [19].

Furthermore, many commercial products, such as toys and games, include speech recognition capabilities [5]. AIBO is a series of robotic pets designed and manufactured by Sony. The name "AIBO" stands for "Artificial Intelligence Robot." In AIBO, voice recognition capabilities are built in, which allows the device to understand verbal commands and respond appropriately to them. It is not just able to react to linguistic cues, but AIBO can also detect and locate the sources of sound, turning dynamically towards a person's voice to determine where it is coming from. A new generation of artificial intelligence and speech recognition technologies has expanded its influence on the toy industry. The first AI-powered doll in market, Hello Barbie uses speech recognition and progressive learning to interact with children [20].

The applications of ASR are not only limited in the field of education & assessment, entertainment, and language learning tools & tutors but equally used in the medical field. Speech interfaces are well established for treating Cleft lip and Palate, Dyslexia, and Autism patients with the ASR system [5, 21]. Dyslexia is a learning disorder, especially a reading disability, related to problems identifying speech sounds and decoding them. The uses of ASR and dialog technology together are helpful for children with Dyslexia. Autism spectrum disorder is emerging and challenging for interactive speech recognition technology [5]. Additionally, the study is focused on infants' pre-speech development. Considering the direct correlation between children's health and pre-speech development, studying pre-speech development is very relevant. Children with Autism disorder can detect earlier, and treatment can start at the earliest. Mood-related infant voice might help caretakers track their children's daily moods and make better decisions [5].

Lee and his team [22], in 2004, worked on children's speech interactive reading program named The Colorado Literacy Tutor (CLT). Improving students' achievement and literacy in public schools is the primary aim of this program based on scientific reading and cognitive theory.

Applications of children's ASR cover various domains, as illustrated in Fig. 1.

3 Challenges in child ASR

Major challenges for the limited number of children's ASR systems are as follows:

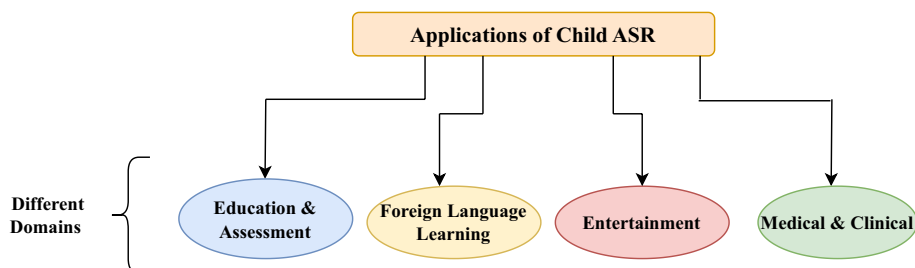


Fig. 1 Applications of Child ASR

3.1 Children's corpus collection [23]

When it comes to data collection, it is found that children's speech data collection is more difficult compared to adults. It is rare for children to concentrate for long periods, and they might hesitate to speak loudly and clearly until they feel comfortable doing so. As a result, children's voices are sometimes recorded in their homes to feel comfortable and relaxed. It takes a long time and is a complex process [23, 24]. Obtaining speech data from pre-schoolers is more difficult due to their inability to comprehend transcripts. To collect speech data from such children, different methods are used. The voice of the child can be recorded after an adult says a word and the child repeats it. The same word can be spoken three times by children, and this method found that children speak the same word with different styles. Further, pictures and animated stories can be used to support recording. Another factor that complicates this process is the variability of children speakers.

3.2 Accessibility of available child corpora [23]

An important reason for the ineffectiveness of children's ASR systems lies in the non-availability of children's corpus [10, 14, 23]. Besides English, children's corpora are published in Swedish, Italian, Dutch, and German as well. Language and age can be considered when analyzing the available data on children's speech. Children's speech corpora are listed in Table 1 (English) and Table 2 (other than English).

The research community has access to a limited number of databases of children's speech from around the globe. These databases have a limited age range for youngsters, and many do not speak English. Furthermore, the diversity of English accents across different countries and backgrounds makes it difficult to include multiple younger children's speech corpora [25]. Furthermore, some datasets are problem specific with restricted tasks, and

Table 1 Children Corpora in English

Speech Corpus	References
TIDIGITS (6 and 15 years)	[23, 30]
CMU kids (6 to 11 years)	[23]
ChIMP Corpus (8—14 years)	[31]
CID Children's Corpus (5 to 17 years)	[32]
OGI Kid's Speech Corpus(5 to 11 years)	[33]
CU Kids's Prompted and Read Speech Corpus (4 to 11 years)	[23]
Tball Corpus (5 and 8 years)	[34]
CU Kid's Read and Summarized Story Corpus (6 to 11 years)	[23, 35]
Providence Corpus (1 to 3 years)	[23]
PF- STAR Children's Speech Corpus (4 to 14 years)	[36, 37]
YOUTH Corpus (8 and 10 years)	[38]
BIRMINGHAM Corpus (4 and 14 years)	[38]
Boulder Learning—MyST Corpus (v0.4.0) (G3—G5)	[39]
AUSKIDS Corpus()	[25]

Table 2 Children's Corpora (other than English)

Language	Corpus	Reference
Hungarian	SPECO corpus (Between 5 and 10 years)	[40]
Multilingual, including Spanish, Russian, German, Danish, Flemish, French, Swedish, Finnish, British English, Mandarin, Dutch, Japanese, Polish, Portuguese, German from Switzerland, Hebrew, American English, American Spanish, Taiwan, and Italian	SpeeCon corpus (Between 8 and 15 years)	[41]
Swedish	PIXIE corpus (ages are not documented)	[23]
German, Swedish and Italian	PF-STAR corpus (Between 4 and 15 years)	[36]
Swedish	NICE corpus (Between 8 and 15 years)	[23]
Italian	ChildIt corpus (Between 7 and 13 years)	[23]
Italian	Tgr-child corpus (Between 8 and 12 years)	[23]
Italian	SponIt corpus (Between 8 and 12 years)	[23]
Japanese	takemaru-kun corpus (There is no documentation of age, but the majority of children are in lower grades)	[42]
Dutch	CHOREC corpus (Between 6 and 12 years)	[43]
Mandarin Chinese	SingaKids-Mandarin (7 to 12 years)	[26]
Kannada (South Indian language)	NITK Kids' Speech Corpus (2 1/2 to 6 1/2 Years)	[44]

are not available publically. Hence, it is essential and has become common for researchers to create their corpora to carry out various research activities in speech [24, 26–29]

3.3 Variabilities of child speakers

The substantial variation in a child's speech often makes their recognition systems less accurate than those of adults. Acoustic and linguistic variabilities are the primary causes of these differences. The speaker's physiology controls these two properties. The acoustics of speech signals are directly influenced by the physical dimensions of the vocal tract and vocal folds, and the control of articulators impacts linguistics in speech. In comparison to adult speech, child speech has significant acoustic and linguistic characteristics [7, 32, 45]. Change in speech characteristics of children due to their rapid growth and at varying rates, is another factor responsible for low recognition of children ASRs. Anatomy and physiology vary significantly when children grow [46]. Consequently, this induces great variations in inter and intra speaker acoustic variabilities in children's speech than in adult language. Furthermore, acoustic differences in children include the pitch, the glottal flow parameters, the average phoneme

duration, the speaking rate, the formant frequencies, and vocal effort; whereas linguistic differences include *correct pronunciation* (usually young children have not learned articulation of specific phonemes properly), the way *children use language* (because of the small vocabulary, children may use certain words that may not be incorporated in adult's speech), *degree of spontaneity*. Unlike adults, children have shorter vocal folds and lower vocal tracts, which results in greater formant frequency and fundamental frequency. The wide distance between the harmonics highlights the increasing fundamental frequency, resulting in poor spectral resolution of spoken words. The non-linear variation in formant frequencies is caused by the variation in vocal tract size [47]. Since the speech production and associated dynamics of children's and adults' speech differ significantly, there exist differences between the speech signal characteristics of adults and children. Following are some important differences between an adult male and a 3-year child, given in Table 3 [7, 48].

3.4 Non suitability of child ASR of one age group with other age group

Age-specific ASRs are not suitable for other age groups [14, 49, 50]. Children aged 2 to 4 years cannot precisely communicate their words and emotions. Children from 5 to 12 years of age can express their feeling using the proper pronunciation of words; however, the collection of speech data from this age group is challenging because the speech data is not publicly available. Another age group is from 6 to 18 years, from which speech data can be collected, but there are many variations in children, especially in males, of this age due to puberty. It has been observed that there is no significant difference in the length of the vocal path of 11-year males and females. However, the formant frequency of females is higher than that of males of the same age. Girls have a steady progression of the vocal tract from adolescence to maturity. The male speaker's pitch drops dramatically from 12 to 15 when they reach puberty. At the age of ten and fifteen years, the male and female formant frequencies separate completely. Male speakers' age is linearly related to vowel-dependent formant frequencies, while female speakers are not.

Young children are also observed to have more spectral and formant variation than adults. Children may have difficulty in coordinating their articulation organs because they are less skilled at it [5, 23, 47, 49]. Ideally, speech recognizers should be trained with similar data to the data they will be recognizing. This ensures the best recognition results. Wilpon and Jacobsen [50] found in 1996 that different acoustic models were developed for children of different ages, and that the generated models had the greatest performance for only the same group of children. The performance of acoustic models established for one age group drops when used for another age group. Large volumes of data are necessary to achieve the most significant recognition results, and gathering such data is problematic for youngsters.

In this section, significant challenges in children's ASR have been identified. Their solutions are also incorporated into subsequent sections of this paper. Challenge 1 given in Section 3.1 and challenge 2 explained in Section 3.2 can be overcome by using data augmentation techniques on children's speech data. Data augmentation and its different methods are explained in Sections 6 and 6.1. Similarly, challenge 3 in Section 3.3 and challenge 4 in Section 3.4 can be controlled using various acoustic mismatch approaches. Acoustic mismatch and its different approaches are discussed in Section 5. Figure 2 shows the relationship between challenges and their possible solution.

Table 3 Differences between children's & adult speech production system [7, 48]:

S. No	Parameters	Child (3 year old)	Adult (Male)	Effects
1.	Height of the head	18.8 cm	23.6 cm	These impact the resonance properties of speech
2.	Length of the vocal tract	10.4 cm	16.9 cm	
3.	Lung vital capacity	940 cm ³	4450 cm ³	The decreased lung capacity of children leads to a greater breathing rate, which enhances the non-stationarity features of their speech
4.	Rate of respiration	24 breaths/min	12 breaths/min	
5.	Length of vocal folds	0.45 cm	1.8 cm	In children's speech, the pitch period is small because their vocal folds are smaller than those of adults, resulting in high vibration
6.	Mass of vocal folds	0.031 g	0.141 g	—
7.	Stiffness of the vocal folds	2.3 * 104 dyne/cm	7.4 *104 dyne/cm	Stiffness in the vocal folds is comparatively more in children. As a result, stronger subglottal pressure maintains faster vibration
8.	Subglottal pressures	12 cm of H ₂ O	18 cm of H ₂ O	Children have a lower speaking rate than adults because of the lower articulation repetition rate
9.	Articulator repetition rate	3 Hz	6 Hz	

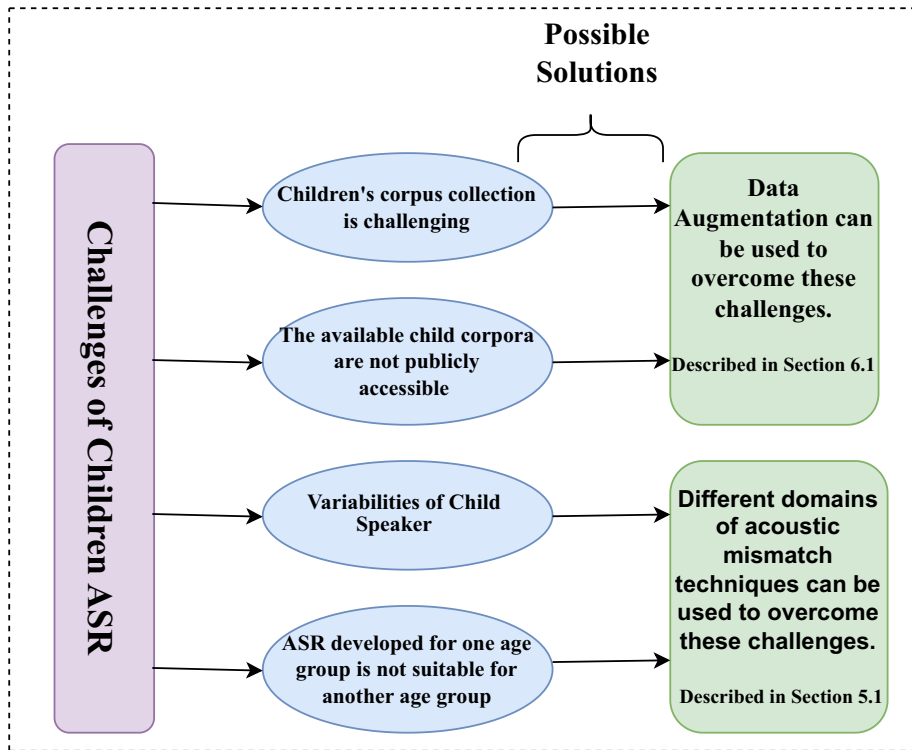


Fig. 2 Challenges of children ASR and their possible solutions

4 Characteristics of children's speech

Speech characteristics of children are broadly classified into the following categories:

- i. Acoustic characteristics
- ii. Linguistic characteristics

4.1 Acoustic characteristics of children's speech

Children's acoustic characteristics change rapidly with age and are also widely different from adults [31]. The anatomical and physiological changes that occur throughout with child's growth are the prime reason for rapid changes in acoustic characteristics. Further, these development changes could generally impact the spectral and temporal properties of the children's speech signals; consequently, inter and intra-acoustic variability are significantly higher in children than in adults. Children become more proficient in articulation with age [47].

Children's speech is characterized by a fundamental frequency (pitch), formant frequency (frequency and bandwidth), and duration (utterance, syllable, segment) when compared with adults' speech. These acoustic characteristics discriminate children's speech from adults' speech. A thorough study of the acoustic measurement pattern indicates the

development of children's articulation and phonological systems. Children have higher fundamental and formant frequencies when compared with adults' speech. A shorter vocal tract and the length of vocal folds are the most critical factors responsible for higher fundamental and formant frequencies [5]. However, children's utterances are usually shorter than those of adults. Further, children usually utter only a few syllables, and the duration of utterances of these syllables is about twice as long as adults. As far as children's vowel frequency is concerned, it is comparable with adults but scaled to higher frequencies. Furthermore, formant bandwidth in children's speech is substantially wider than in adults. Various research groups are working on the above mentioned acoustic characteristics of children's speech [32, 46, 51–54]. The changes in formant and fundamental frequency, which are dependent on the age of the children ranging between 3 to 13, were reported by [5, 55]. Children's recognition performance, though it gradually improves with age, tends to be lower than that reported for adults, despite a sufficient amount of age-specific training data available [22, 42, 50, 56–58]. As children grow during the early years of their life, the length of their vocal folds increases because of the increase in thickness of their vocal folds. The formant values of a child's speech increased non-linearly, and these formants are located at high scale [7, 48]. The same has been observed in experimental research done by [10, 32, 47, 59]. Further, it has also been observed by various researchers that the increase in the duration of utterance reduces their speaking rate, compared with adults, which is higher in the case of adults. The phoneme duration analysis might be affected by mispronunciations and time alignment issues [47]. The longer phoneme duration indicates the growth of lungs in children, which in turn increases the ability of children to control air-flow to glottal and supraglottal constrictions [48]. The difference between consonant–vowel transitions of children and adults has been observed by [5, 60]. Shorter transition duration and longer spectral differences between consonant & vowel have been found in children's speech for consonant–vowel pairs compared to adults' speech. It is also observed that the values of acoustic parameters such as formant, pitch, and duration decrease as the children grow. Further, these values reach the adult range at the age of 13–14 years [32, 47]. Formant frequencies are affected by radiation, glottal resistance, wall resistance, viscosity, and heat conduction. The relation between these factors and formant frequencies is given in Table 4 [7, 48].

Further, Bg and Br are the principal influencers of bandwidth, with Bg strongly contributing to the bandwidth of the first formant and Br strongly contributing to the bandwidth of the higher formants. As a result of children's larger vocal tracts and higher formant frequencies, children contribute larger bandwidth that contributions to radiation effects than adults. Additionally, children have a greater bandwidth contribution due to the glottal resistance effect. Because of glottal resistance, the bandwidth of the first formant is 66 Hz and 53 Hz for children and adults, respectively, which is higher for children [7, 48]. Three main parameters determine the glottal flow of a speech signal, namely open quotient (OQ),

Table 4 Relationship between various factors and formant frequency

S.No	Factor	Relation with Formant Frequency
1	Radiation (Br)	$Br \propto (\text{Formant Frequency})^2$
2	Glottal Resistance (Bg)	$Bg \propto (\text{Area of Glottis})^2$
3	Viscosity (Bv)	$Bv \propto (\text{Formant Frequency})^2$
4	Heat Conduction (Bh)	$Bh \propto (\text{Formant Frequency})^2$
5	Wall Resistance (Bw)	$Bw \propto 1/(\text{Formant Frequency})^2$

return quotient (RQ), and speed quotient (SQ). The shape of the glottal pulse and glottal flow parameters affect the long-term shape (spectral tilt) of the speech power spectrum [61, 62]. As a consequence, large OQs usually indicate that the lowest few harmonics are more prominent in the spectrum of the source. With a large RQ, the higher frequencies are attenuated more, and the steepness of the source spectrum is affected. Children and adults have also been found to have different glottal flow parameters (OQ, RQ, and SQ) [63, 64]. Therefore, it is found that due to these differences in the glottal flow parameters, there is also a difference in breathiness in speech [65–67]. Furthermore, there is a higher level of breathiness in young children's speech than in adults' speech [56].

To summarize, at the end, the higher formant values, high pitch values, and slow speaking rates are perceptually relevant sources of acoustic mismatch between children's and adults' speech. Performance degradation is due to acoustic mismatch and variability. Mainly speaker normalization, model adaptation, and signal domain models are used to reduce variability and constrain the acoustic space of children's speakers [68].

4.2 Linguistic characteristics of children's speech

Linguistic mismatch in children's speech plays a vital role along with acoustic mismatch. Since a child has not fully developed the knowledge of a language, he/she cannot use the rules of that language, which leads to linguistic variabilities and affects ASR's performance. The ASR's performance is further affected by several linguistic differences. Hence, it is essential to explore linguistic differences between a child's and an adult's speech. This section discusses the important reasons or factors that are considered to be the main reasons for linguistic mismatch.

Lacks of control over the articulators, smaller vocabulary, and of course, mispronunciations are primary factors for linguistic variabilities in the case of children. Children cannot learn articulation of specific phonemes in the age range of 5–6 years. This lead to many other issues in children's speech, like speech disfluencies, false-start, and extraneous speech [31, 53]. Further, in 8–10 years old children frequency of occurrence of mispronunciation is almost twice as for children of the age group 11 to 14 years. Children generally use fewer words per utterance when conveying their messages or thoughts as they know very few words and have a small vocabulary than adults. However, children made more grammatical mistakes than adults for spontaneous speech, and low vowel duration and high speaking rate were also observed for spontaneous speech rather than read speech [52]. Children may not have fully learned the right inflectional forms of some words, particularly those that are exceptions to conventional norms. As a result, their utterances occasionally contain terms that are not seen in adult sentences. Hesitation is another crucial factor that can be seen in child speech. It has been observed that girls in the age group 10 to 11 years have more hesitation compared to boys of the same age group. It is difficult to understand factors for hesitation, but societal factors might be attributed to this rather than linguistic skills. Furthermore, it has been seen in the girl's age group of 12–14 years that their average sentence length increased, and this led to an increase in linguistic variability. In the above-mentioned age group, girls show more language development than boys of the same age group [51].

In a nutshell, with age, linguistic variability in children's speech diminishes. Speech disfluencies in children decrease as children grow older. Usually, children get ready to speak like an adult from twelve to thirteen years of age, a little faster for boys than for girls. As youngsters become more senior, their capacity to communicate effectively through

language develops. Children's speech differs significantly from adult speech concerning the variability of acoustic and linguistic correlates and absolute values.

5 Approaches in child ASR

Acoustic mismatch and variability are identified as the major contributors to the performance degradation of children ASR. Speaker normalization, age-dependent models, and model adaptation are used to reduce variability and constrain the acoustic space of children speakers. Various approaches have been explored in literature so far to address different acoustic and linguistic mismatches for improving children's ASR on adults' speech trained models. These approaches are briefly reviewed in this section:

5.1 Acoustic mismatch

When data is acquired from adults, the acoustic model is trained with that data, and testing is also done with adult data; this is referred to as a "*matched condition*" because the variables in training and testing are the same. A "*mismatched situation*" occurs when ASR is trained on adults' speech data and evaluated/tested on children's speech data, or vice versa. In mismatched conditions, the variabilities are different during training and testing. The mismatched ASR refers to training the acoustic model using adult and children's speech data and testing with only children's speech data. When the variabilities present in the training or testing speech data are not the same, it's called mismatched ASR [5–7]. There are two approaches to develop ASR system for children. One approach is to collect children's speech data separately and build the whole system using only children's speech. However, due to a dearth of publicly available children's speech data, constructing a well-trained ASR framework is challenging. The second approach is to build the system using adult speech data. The lack of children's speech data could be compensated by blending adult and child speech for training. Additionally, there are several applications that require the recognition of both adult and child speeches. However, poor recognition performance has been documented in literature for such ASR systems [1, 5, 6, 14–17, 21, 23, 50, 68–71].

Acoustic mismatch and variability have been recognized as the primary causes of performance deterioration in the development of high-performance ASR applications for youngsters. Spectral and temporal variabilities make acoustic modeling difficult. Age-dependent models, model adaption, and speaker normalization are used to minimize the effect of acoustic variability of children speakers [68]. In order to improve acoustic differences between adults and children's speech, different methods have been explored in literature. Different methods reported in the literature may be represented by three distinct groups based on the field in which different acoustic sources of mismatch are addressed; feature domain, signal domain, and model domain [5, 72].

5.1.1 Feature domain approaches

Normalizing techniques are used in the feature domain approaches to reduce the mismatch of recognition characteristics like cepstral coefficients [72]. Numerous researchers worked on the domain of features to reduce the mismatch between child speech and adult speech and found significant improvement in the performance of child ASR systems. In [69], the authors designed an adaption method for neural network-based recognizers. Each speaker's

Bark offset is used in the suggested method to execute a speaker-dependent warping of the frequency scale. Adults' speech-trained models showed 4.2% and 3.5% improvement in children's ASR performance using single-digit and seven-digit utterances for adaptation, respectively. In this study, Bark shift is examined in voiced and unvoiced speech. But in reality, unvoiced speech is less impacted by the vocal tract, so it is not necessary to include unvoiced speech. This can be examined in the future. Further, this research was carried out on unsupervised adaptation, although it needs to be seen if supervised adaptation would provide greater results. In 1996, Wilpon and Jacobsen [50] worked on children's speech recognition performance on child corpus collected over the telephone and a connected digit recognizer. They found that the Word Error Rate (WER) of children's speech is more than 100% higher than that for adults' speech. The acoustic models were trained using an equal amount of speech data from the age group of 8 to 80 years [50]. In 1997, Potamianos and co-workers [68] proposed linear warping speaker normalization, spectral shaping adaptation, and age-dependent acoustic modeling on mismatched data and found improved recognition performance up to 55% for children speakers, using a single utterance for adaptation. By adjusting the acoustic model based on the age of the speaker, the word error rate is further reduced by 10%. The acoustic mismatch between children and adults is reduced by the use of a piece-wise linear and phoneme-dependent frequency warping algorithm. The research was done for matched and mismatched conditions by Giuliani & Gerosa [73]. The authors worked on children's speech recognition by taking into consideration the phoneme recognition task. It was found that under matched conditions, for both children and adults, the phoneme's recognition accuracy was 77.30 percent and 79.43 percent, respectively. The improvement has been shown by applying Vocal Tract Length Normalization (VTLN) for both under matched conditions and for mismatched conditions. For matched conditions, VTLN gave 10.5% and 5.3% error rate reductions for adults and children, respectively. Additionally, under mismatched conditions, the performance improved up to 66.43%, which was 58.11%, with the baseline system. According to Elenius & Blomberg [74], in 2005, children's speech differs acoustically in terms of lexical and grammatical levels. Adaptation and normalization might produce an improved performance. Further, mismatched conditions result in lower performance than matched conditions. By adapting and normalizing the vocal tract length of adult models, poor performance was significantly improved, but not to the same extent as training on children. As compared to dedicated models for each group of speakers, a combined model trained on both adults and children performed almost well. In order to improve the performance of piece-wise linear VTLN algorithms, lower warp factors are needed for children's speech. A modified model would likely perform better than a normalized model for both linear and piece-wise linear algorithms. A phoneme-based model, even with a small vocabulary, performed better than a word-based model. Authors have shown that Maximum Likelihood Linear Regression (MLLR) performed better than VTLN, reducing the word error rate by 19.3%. Further, it was reported that the Maximum a posteriori (MAP) and MLLR are significantly more effective than piecewise linear VTLN. In the case of age-dependent models created by modifying the child model, there was no discernible improvement. Cui & Alwan [75], in 2006, investigated on discrete frequency domain and followed by the relationship between the front-end linear frequency warping and model transformation back-end. Various feature extraction schemes, such as cepstra with Mel-scale warping, cepstra with and without Mel-scale warping, and cepstra with Mel-warped triangular filter banks, were used to investigate the above-mentioned relationship. Children's speech is adapted using adult-trained acoustic models, and there is a limited amount of adaptation data available. Peak alignment schemes obtained the best results as compared to traditional VTLNs and MLLRs on different amounts of adaptation

data. On average, with the widely used Mel Frequency Cepstral Coefficient (MFCC) feature with Mel-warped triangular filter banks, speaker-specific F3 alignment outperformed VTLN by 33%, MLLR by 54% with limited adaptation data. H^oonig and co-authors [76] have worked on the two most common and popular feature extraction techniques; MFCC and Perceptual Linear Prediction (PLP). Acoustic features were developed by combining the advantages of MFCC and PLP. The authors revised the processing steps of PLP. They improved the filter bank, the equal-loudness pre-emphasis, and the input for the linear prediction in traditional PLP, and noticeable improvement was reported. When compared with conventional PLP, revised PLP reduced WER by 5.7% rel. improvement for ChildIt corpus and by 4.4% rel. improvement for HUB4. Compared to the MFCC and PLP baseline systems, WER has improved by 3.9% and 4.4%, respectively. According to MLLR, the revised PLP with 257 filters performs in both cases 4.8% better than the baseline MFCC and PLP. This new variant performs better in noisy conditions than MFCC and conventional PLP [76]. Gerosa and his team [47], in 2007, analyzed children's and adults' speech by giving focus on temporal and spectral variability, phoneme duration, and the effects of vocal tract length. Several methods for speaker-adaptive acoustic models have been explored by the authors. The use of VTLN significantly improves the performance of children's ASR systems. Apart from VTLN the impact of Constrained MLLR Speaker Normalization (CMLSN) and Speaker Adaptive Training (SAT) on child ASR has also been explored, and it was reported that these methods improve the performance of child ASR significantly in mismatched conditions. The authors claimed that CMLSN method outperforms other methods used in this work. Relative WER of 8.6% and 28.0%, w.r.t. baseline system, was obtained by using cascading of VTLN and CMLSN. A recent advancement in acoustic and language modeling is presented in [77] to improve the recognition performance of children when they read aloud in digital books for the first time. A 54% reduction was reported in word error rates following the addition of Vocal Tract Length Normalization (VTLN), Speaker-Adaptive Training (SAT), and iterative unsupervised structural maximum a posteriori linear regression (SMAPLR) adaptation. On the story read aloud task, the current WER is 7.6%, while on the summary recognition task, it is 32.2%. Yeung and co-workers [78] explored a new technique by combining data augmentation and feature normalization, which is based on formants and fundamental frequency relation. Data augmentation and feature normalization are both implemented as frequency shifts within the Mel domain. An adapting BLSTM-based acoustic model is trained using adult speech and tested using child speech. The proposed technique showed an improved relative WER by 19.3%. These methods will be evaluated on both child and adult ASR systems where there is more variation in age and gender between speakers. In addition to conversational speech, educational applications, and clinical applications, apply these systems to other child ASR applications as well.

In [79] mismatches, due to adult and children's speech, in formant frequency were minimized by adjusting the formants of children's speech to more closely align with the formants of adults' speech. This is obtained by warping the linear prediction (LP) spectrum derived from children's speech. The adults' and children data set were taken from WSJCAM0 and PFSTAR respectively. For narrow-band speech, the suggested method improved DNN and TDNN acoustic models by 24% and 11%, respectively. For wide-band speech, the technique improves acoustic models by 27% and 13%, respectively. In [80], prosodic features were combined with MFCC and PLPCC in order to see their effectiveness. A significant improvement has been noted with this combination. The system performance was further enhanced by the use of low-rank feature projection. The research reported in this article [81] demonstrates the effectiveness of suggested acoustic features

and out-of-domain data augmentation techniques for zero-resource children's speech recognition tasks in both clean and noisy conditions. The authors investigated the impact of duration and format modification-based data augmentation in this study and they found that the performance of the recognition system has been improved. New filter-based discriminative auto encoder (f-DcAE) architecture for acoustic modeling is proposed in [82]. The authors [83] explored pitch-adaptive signal processing to extract front-end features for children's ASR. Before calculating the frequency, an adaptive-liftering approach is presented for smoothing magnitude spectra. Pitch-robust and filterbank features are derived using an adaptive-liftering approach. Using the DNN-based acoustic modeling paradigm, the proposed pitch-adaptive features are demonstrated to be effective. Children's mismatched ASR can be effectively addressed with both kinds of pitch-adaptive front-end features. Various feature domain approaches are given in Table 5.

5.1.2 Signal domain approaches

To minimize noise components in input noisy signals, the signal domain method uses single-channel speech improvement algorithms as a preprocessor for ASR. The design of signal models has traditionally been influenced by the robustness of the models versus background acoustic noise. Speech recognition technologies today are highly sophisticated as the ASRs themselves contribute to noise robustness. Verification of speaker, speech recognition, and speech and language-related research still focuses on signal models. There may be several types of signal models that are optimal for certain types of applications but not necessarily for others [88]. Cabral & Oliveira [89], in 2005, used LP PSOLA for their research because they were interested in the characteristics of the glottal signal for which the linear prediction residual is an approximate estimate of its derivatives. However, a small modification is done in LP PSOLA; a time-scaling transformation is proposed to keep the shape of the residual waveform within the period. The overlapping interval can be altered by time-scaling the short-term signals while retaining the updated signal's energy balance. Pitch synchronization is done by utilizing a pitch marking technique to produce an approximation of the glottal closure instants for vocal speech. This modified LP PSOLA is a Pitch Synchronous Time Scaling (PSTS) algorithm of the linear prediction residual. In comparison to previous time-domain approaches, this method provides more flexibility for prosody modification and voice quality change of speech with very less increase in complexity. Das and his team [70], in 1998, observed the benefits of frequency warping on a recognizer where acoustic and language models were based on speech data collected from children. This recognition system performs with a word recognition error rate of 10.46%. In their research, the authors used two types of warping: 'speaker dependent (SD)' and 'speaker independent (SI)' warping. In SD warp, the warp factor is optimized independently for each speaker, whereas in SI, a uniform warp factor is optimized. Overall 11 speakers are used. The authors further suggested the same technique; if applied to large-size vocabulary, the performance of the system can be improved.

To increase the speech recognition performance for children Stemmer and his co-works [49] evaluated two new ways. i) Non-linear VTLN and ii) PSOLA-based normalization of speaking rate. The two methods lead to a reduction in word error rates for a speech recognizer that is trained in adults' speech. The authors, however, were not able to find a better "Mel-scale for children" that would also improve recognition performance when the recognizer is already trained on children's speech. the focus of the research was on the problem of adapting the acoustic frontend to achieve better performance of mismatched

Table 5 Feature Domain approaches

Reference	Language/Corpus/ Data Set	Children/ Age Group (in years)/ Grade	Feature Extraction Technique	Acoustic Modelling	Feature/ Algorithm	Accuracy/ WER	Remarks
[47]	Three corpora were used: 1. Italian, ChildIt, 2. Italian, SpontIt, 3. American English, CID,	171 speakers 7 to 13 21 speakers 8 to 12 436 speakers 5 to 18	MFCC	HMM	VTLN	—	Highest recognition results were obtained by cascading the VTLN with the CMLSN method
[50]	Danish Rafael0 telephone speech database	306 speakers 8 to 18	—	HMM	—	WER is more than 100%	The acoustic models were trained using speech data in the age group of 8 to 80 years. The WER of children's speech > 100%, than that for adults' speech
[68]	DgtII ("(\CHLD HMM") Subw II Dgt III Comm I Comm II	10 to 17 years 10 to 17 years 6 to 7 years 6 to 7 years 10 to 17 years	MFCC	HMM	frequency warping	55%	-Combining spectral shaping and frequency warping was used to explore speaker normalization -Bi-parametric and phone-dependent warping functions were also investigated as an alternative to linear frequency warping
[69]	TIDIGITS corpus	51 children	PLP	LPC cepstral analysis	—	4.2% and 3.5%	-ASR performance was evaluated using single-digit and seven-digit utterances for adaptation -Mismatched ASR

Table 5 (continued)

Reference	Language/Corpus/ Data Set	Children/ Age Group (in years)/ Grade	Feature Extraction Technique	Acoustic Modelling	Feature/ Algorithm	Accuracy/ WER	Remarks
[73]	Two task-inde- pendent speech databases (Consisted of Italian read sentences.)	7 to 13 years	MFCC	HMM	VTLN	66.43%	-For matched conditions, VTLN gave 5.3% and 10.5% WER reduction for children and adults, respectively -For mismatched condi- tions, the performance improved up to 66.43% unit accuracy with VTLN Even for small vocabular- ies, Phoneme-based models outperformed the word based models
[74]	Swedish, PF-STAR,	198 speakers 4 to 8	MFCC	HMM, GMM		—	With limited adaptation data, the speaker-specific F3 alignment scored bet- ter than VTLN, MLLR, and Mel-warped triangu- lar filter banks
[75]	TIDIGITS database	10 speakers 6 to 15	MFCC	HMM, GMM		33%	-Relative improvements in WER compared to the baseline systems of MFCC and PLP are 3.9% and 4.4%, respectively -MLLR shows that the revised PLP with 257 filters performs 4.8% bet- ter than MFCC and PLP baselines
[76]	Childt and HUB4	7 to 13	MFCC PLP RPLP	HMM,GMM		5.7%	

Table 5 (continued)

Reference	Language/Corpus/ Data Set	Children/ Age Group (in years)/ Grade	Feature Extraction Technique	Acoustic Modelling	Feature/ Algorithm	Accuracy/ WER	Remarks
[77]	read data, 106 children	Grade 3 to grade 5	MFCC PMVDR	HMM		_____	Real-time system imple- mented that operates on small pipelined audio segments (250 ms on average)
[84]	English, TI-digits, isolated digits and digit strings,	51 speakers 7 to 14	MFCC	HMM		more than 50% improvement has been shown	WER (under mismatched condition) with trans- formed adult models com- pared to non-transformed case, decreased by more than a factor of two
[85]	The speech of English and Italian children was collected and analyzed in two databases a) Native English Children b) Non-native Italian Children	a) 7 to 12 b) 10 to 11	Mel frequency cepstral	HMM, GMM		Italian children made 100%- 600% more word errors than English children of the same age	The authors concluded that there exist orderly differ- ences between non-native and native speech
[78]	OGI Kids' Speech Corpus, 100 speakers CMU Kids Corpus, 76 speakers	kindergarten to 10th grade 1st and 3rd grade	MFCC	3-layer BLSTM network	Fundamental Frequency	19.3% relative improvement Less significance Improvement	A novel technique for child ASR is proposed using both feature normalization and data augmentation methods. These tech- niques are implemented as a frequency shifts in the Mel domain

Table 5 (continued)

Reference	Language/Corpus/ Data Set	Children/ Age Group (in years)/ Grade	Feature Extraction Technique	Acoustic Modelling	Feature/ Algorithm	Accuracy/ WER	Remarks
[86]	British English PFSTAR	122 speakers 3–14 Years	PLPCC, SMAC PN-SMAC	GMM,DNN,SGMM	Pitch	In the narrowband condition, adults’ speech-trained ASR systems produced a WER of 52.38%	-A novel front-end speech parameterization approach is proposed -In terms of pitch-robust- ness, the proposed acous- tic features improve the existing SMAC technique
[87]	OGI Kids’ Speech Corpus,	100 speakers kindergarten to 10th grade	MFCC	DNN-HMM	fundamental frequency (f ₀), VTLN, F3-based, and SGR-based normalization	<i>f</i> ₀ performs better than others	Spectral mismatch between kindergarten and older child speech are reduced using fundamental frequency (f ₀)-based normalization
[79]	British English, PF_STAR	122 speakers, 4 and 13 years	MFCC	GMM-HMM, DNN-HMM, TDNN	Formant Frequency	RI a) For narrowband speech: 24% (DNN) & 11% (TDNN) b) For wide- band speech: 27%(DNN) & 11%(TDNN)	The WER for mismatched system reduced using LP based formant modifica- tion which could further be reduced by integrating above with VTLN and SRA
[80]	British English, PF_STAR	122 speakers, 4 and 13 years	MFCC,PLPCC	GMM-HMM, DNN-HMM,	Prosody Features: loudness (LD), voice-probability (VP) and voice- intensity (VI)	Upon merging MFCC (or PLPCC) with prosodic fea- tures, observed RI of 16% (14%)	Low rank feature projection used to increase system’s performance

Table 5 (continued)

Reference	Language/Corpus/ Data Set	Children/ Age Group (in years)/ Grade	Feature Extraction Technique	Acoustic Modelling	Feature/ Algorithm	Accuracy/ WER	Remarks
[81]	PF-STAR	122 speakers, 4 and 13 years	MFCC	DNN-HMM	Formant Frequency	—	The authors investigated the impact of duration and format modification-based data augmentation and found the improved performance of the recognition system
[82]	CMU Kids corpus	151 speakers 6–11 years	MFCC	filter-based discriminative autoencoder (f-DcAE), TDNN	Pitch Features	7.8% relative WER reduction compared to the baseline system	f-DcAE models overtake the counterpart baseline models without using the autoencoder architecture and the filtering mechanism
[83]	British English, PF_STAR	60 speakers 4–14 years	MFCC	DNN-HMM	Pitch Features	—	Using DNN-based acoustic modeling, authors demonstrate the effectiveness of the proposed pitch adaptive feature
[78]	OGI Kids' Speech Corpus	151 speakers kindergarten to 10th grade	MFCC	a BLSTM-based acoustic model	fundamental frequency	RI of 19/3%	A technique for child ASR is introduced which incorporates feature normalization as well as data augmentation methods based on a relationship between formants and fundamental frequency (fo)

ASR. Potamianos and co-workers [68] proposed linear warping speaker normalization, spectral shaping adaptation, and age-dependent acoustic modeling on mismatched data and found improved recognition performance up to 55% for children speakers, using a single utterance for adaptation. By adjusting the acoustic model based on the age of the speaker, the word error rate is further reduced by 10%. The acoustic mismatch between children and adults is reduced by the use of a piece-wise linear and phoneme-dependent frequency warping algorithm. On the other hand, the purpose of the study done by D'Arcy & Russell [90] is to measure human recognition performance for children's speech. An analysis of the same children's speech data is conducted by comparing human and machine recognition performance. According to this study, humans are able to recognize children's speech with similar effects depending on their age and bandwidth as automatic systems. Gustafson & Sjolander [91] transformed speech signals using two transformations techniques, and then it is sent to the recognizers. Two transformation techniques that are used are *i*) phase vocoder algorithm and *ii*) TD-PSOLA; in this method, some form of formant scaling possibility is also provided by including linear compression of the spectral envelope of each window. For this research, Nuance recognizer, a custom commercial application, is being used for the spoken dialogue system. Reasonably, high acoustic quality is achieved using transformations provided by both methods. The authors suggested that a general approach could be developed to determine when to apply transformation methods from acoustic signals. Umesh and his team [92], in 2004, worked on front-end signal processing. The researchers investigated speaker normalization based on maximum likelihood. Rather than assuming that the whole magnitude spectrum (including pitch harmonics) is scaled, the authors claimed any two speakers' spectral envelopes should be thought of as linearly scaled versions of each other. Implementation of the proposed model on digit recognition tasks shows improvement in adults and children by about 4% and 7%, respectively.

[93] Demonstrated a robust feature extraction methodology based on optimized filtering to address noise issues in children's speech. A comparative analysis was conducted by employing MFCC, LPCC, RASTA-PLP, and GFCC feature extraction techniques to compare Kalman and Wiener filtering techniques. A study has been carried out on Punjabi-language speech data collected from children. The Line Spectral Pairs (LSP) frequency variances have been examined for speech uttered by 10 adults and 10 children [94] and clearly illustrate age-related differences in lower-dimensional LSP frequencies, specifically in the first through third LSPs. Children's speech has lower-dimensional LSP frequencies than adult speech. The mean and variance of the LSPs for each vowel highlight the significance of low-dimensional LSPs as significant speech features. Various signal domain approaches are given in Table 6.

5.1.3 Model domain approaches

A model domain approach is a powerful tool for reducing the mismatch between statistically trained recognition models by means of adaptation techniques such as MLLT (maximum likelihood linear transform), MAP (maximum a posteriori), etc. ASR systems have to adjust their models in order to compensate for mismatches between their test and training data. However, model adaptation is a very challenging and complex task because in it large, complex models need to modify without explicit supervision using a limited amount of target data. An adaptive acoustic model created with MAP optimally uses the prior information obtained from the clean-trained acoustical model and the posterior information obtained from noisy observations. Another important feature of MAP is that only models

Table 6 Signal domain approaches

Reference	Language/Corpus/ Data Set	Children/ Age Group (in years)	Feature Extraction Technique	Acoustic Modelling	Signal Processing Parameter	Technique Used	Accuracy/WER	Remarks
[70]	1st acoustic data- base: KidCC 400 sentences	5 males and 6 females	—	kidAM11	Frequency	Frequency Warping	WER is 10.46%	-The benefit of frequency warping was observed
		From 8 to 13 years		kidAM22				-Language model substantially improves the recognition performance
	2nd acoustic data- base: KidAM 27,650 word	395 children From 5 to 10 years		kidAM22				
[49]	Language Model: kidLM	—						
	240,000 words							
	Testing Database: kidTEST	6 Males & 6 Females						
[71]	Language: German Corpus: 3.5 h	Every child was of 6 years	MFCC+ VTLN	—	Speaking rate	Non-linear VTLN and speaking rate normalization with the Pitch Synchronous Overlap and Add (PSOLA) algo- rithm	For non-linear VTLN optimi- zation: WER is 25.9%	A non-linear VTLN and a PSOLA algorithm are intro- duced to improve children's speech recognition
		62 speakers 10 to 12 years						
	Native Language: German and Swedish	150 speakers 4 to 15 years	—	HMM	—	—	—	Child speech characteristics (age and bandwidth) have a greater effect on ASR accu- racy than ASR technology artifacts
[71]	Non-Native Lan- guage: English							
	PF STAR							

Table 6 (continued)

Reference	Language/Corpus/ Data Set	Children/ Age Group (in years)	Feature Extraction Technique	Acoustic Modelling	Signal Processing Parameter	Technique Used	Accuracy/WER	Remarks
[92]	OGI corpus	6 to 18 years	MFCC	HMM	spectral envelopes	weighted overlapped spectral averaging (WOSA)	7% WER	On a digit recognition task, adults and children performance enhanced by 4% and 7%, respectively, by implementing the proposed model
[93]	Own collected speech corpus in Punjabi language	39 speakers	MFCC, LPCC, RASTA-PLP and GFCC	DNN-HMM	Speech signal	Kalman and Wiener filter	Overall Improvement of 16.13%	Under noisy conditions, the Kalman filtering and Wiener filtering results obtained with synthetic data proved more advantageous for the children's speech corpus, with overall improvements of 24.55% and 30.65%, respectively
[94]	JWC child database	24 speakers 10 or 11 years	Conjugate Structure-Algebraic Code Excited Linear Prediction (CS-ACELP)	DTW	Line Spectral Pairs	—	—	Analysis of the LSP frequencies of the speech data concluded that the main differences between children and adults' speeches are from the frequencies of lower LSP parameters
[95]	PF STAR	150 speakers 4 to 15 years	PACT-MFCC NUSS-MFCC MFC	GMM HMM LSTM	Pitch	spectral smoothing	14.64% with LSTM	To mitigate the acoustic mismatch caused by pitch differences, the authors proposed a non-uniform spectral smoothing algorithm

representing the adaptation data are adjusted [74, 88]. On the other hand, MLLR uses the parameter sharing strategy to achieve good adaptation performance, in which similar models are tied together, and their parameters are adjusted simultaneously. To optimize the likelihood of adaptation data, MLLR adapts models by estimating a linear transformation of parameters. In this method, classes of models can share common transforms. A benefit of this approach is that it allows models to be adapted that have not been observed in the adaptation data. It should, however, be taken into consideration that models which cannot be adapted in a similar fashion must not be clustered together [74, 88]. Further, the distribution of adaptation data should be similar to what is encountered during testing in order for an adaptation algorithm to be effective. In order to fulfill this requirement for acoustic adaptation, a speech segment is typically used as a basis for the adaptation data (i.e., the same speaker, accent, domain, or acoustic environment) [96, 97].

Giuliani and co-authors [98], investigated a speaker adaptive acoustic modeling by using speaker normalization methods. Among these normalization methods, one is novel speaker normalization, and the other one is famous VTLN normalization. The proposed novel normalization method acquires speaker-specific affine transformation estimated through constrained MLLR. Experiments were conducted for matched and unmatched training and testing conditions on two corpora on adults' speech and children's speech. The obtained results confirmed the effectiveness of the proposed method to compensate for the inter-speaker variability. The authors claimed that the proposed method also performs better than VTLN method. Further, the proposed method performs better than MLLR under matching training and testing acoustic conditions. The proposed normalization method can be used for various applications where Microphones, transmission channels, etc., induce acoustic variabilities.

Elenius & Blomberg, [74] worked on different adaption techniques on a set of 50 speakers. These speakers were divided into 5 categories, one for every age group between 4 and 8 years. Each group consists of 6 children. Height adaption was also used, and grouping the adaption divided into 5 height categories with similar speaker counts. Along with height and age, general children adaptation was also performed and for this 60 utterances were chosen from the adaptation set, randomly. This was done to match the size of the age dependent adaptation data. Phoneme based models and word-based model were compare too. The experimental results showed i) All age groups were significantly less affected by word errors with MLLR than with VTLN. ii) MAP and MLLR performed better than piece-wise linear VTLN. They were also better at adapting to children in general than to specific age groups or heights, despite the fact that a similar number of utterances were used in each case. A word error rate of 25.7% was found for training directly on the adaptation set. iii) Further, it was also found that phoneme-based model yielded better results than a word-based model, even for small vocabularies. Hagen and his team [99], worked on children's speech interactive reading program named as The Colorado Literacy Tutor (CLT). Based on cognitive theory and scientific reading, this program aims to improve literacy and students' achievement in public schools. Authors explored unsupervised MAPLR adaptation along with VTLN when applied on spontaneously spoken story summaries. The obtained results have shown the WER of 42.6%. CLT uses the SONIC speech recognition system from the University of Colorado. Cosi and Pellom [100], also worked on the CLT and investigated the combination of VTLN, structural MAP linear regression (SMAPLR) and speaker adapted training (SAT). The phonetic recognition error rate was reduced from 21.5% to 21.0% for the 40-unit system and from 16.4% to 16.0% for the 33-unit system. Eventually, kids reach adult-level skills around 12–13 years of age (a little earlier for boys). When

it comes to hesitations, the age trend reverses [101] Speaker-adapted acoustic models were trained on a mixture of adult and child speech, ensuring recognition performance was equal to group-specific models for adults and children. In contrast to adult utterers, children uttering the same text has a WER of 10.2% compared to 8.2% for adults.

Shahnawazuddin & Sinha [102], in 2015, worked on fast acoustic adaptation based model and interpolation was applied to an online children's ASR. In such tasks where very few parameters need to be computed, fast adaptation techniques are found to be more suitable. It is possible to simulate an online ASR task by conducting unsupervised utterance-specific adaptation experiments. This could be achieved using an acoustic model interpolation and a low-rank feature projection. The models were interpolated based on eigenvoices (EV), reference speaker weights (RSW), and sparse representations (SR). In addition to reducing latency, the proposed approach ensures a low storage requirement overall. In terms of performance, EV and SR-based approaches are almost identical.

Whisper, a new state-of-the-art E2E supervised encoder-decoder transformer-based ASR model, has been investigated in [103]. Multilingual ASR models based on extensively annotated adult speech datasets were used in Whisper development. To enhance the performance of the child ASR systems, the authors examined the adaptability of such models to the child ASR system. Furthermore, the authors compare child-adapted Whisper versions with fine-tuned self-supervised models like wav2vec2. To enhance the robustness of ASR for children's speech, the authors of [104] employed transfer learning on a transformer model pre-trained with adult speech and introduced the source-filter warping data augmentation strategy. To mitigate the domain gap between adult and children's speech data augmentation technique based on the source-filter speech model presented initially. Employing this model for the ASR task involves augmenting adult data with the suggested source-filter warping strategy and a restricted amount of in-domain children's speech. An official test set of PF-STAR British English Children's Speech achieved a 4.86% WER using this approach. The study [105] offers a comprehensive evaluation of automatic children's speech recognition, conducted through an empirical examination of modern state-of-the-art end-to-end speech recognition systems. A comparison of the hybrid TDNN-F HMM, and the E2E architecture on the speech of children was conducted in this study. For children's speech recognition, [106] compares the conventional cepstral feature-based ASR approach with a CNN-based end-to-end acoustic modeling method that simultaneously learns relative features and a phoneme classifier from raw speech. A CNN-based end-to-end acoustic model outperforms systems using standard features such as MFCCs in the PF-STAR corpus. There was a reported 11.99% WER in the PF-STAR corpus, which is the highest reported in the corpus. Gerosa and his co-workers [107] proposed speaker adaptive modeling that aims to reduce acoustic variations generated by different characteristics of each training and testing speaker. Authors achieved speaker adaptive modelling through VTLN, SAT and constrained MLLR based normalization (CMLSN) and additionally from cascading the VTLN method with the SAT and the CMLSN methods. The obtained results show that the cascading VTLN method with SAT and CMLSN gave best results than others. Table 7 summarizes various model domain approaches.

This section compares the various domains of acoustic mismatch, namely the feature domain, the signal domain, and the model domain. It has been observed that some researchers use traditional architectures (HMM) for their implementation, some use hybrid methods (DNN-HMM, GMM-HMM), and some mention End-to-End (E2E) techniques.

It is worthwhile to compare End-to-End (E2E) models with traditional and hybrid models. This can assist readers/researchers in selecting the optimum architecture for their research work. In this paragraph, E2E models are compared first with traditional models,

Table 7 Model domain approaches

Reference	Language/ Corpus/ Data Set	Children Age Group (in years)/ Grade	Feature Extrac- tion Technique	Acoustic Modelling	Model Adaptation Algorithm	Accuracy/ WER	Remarks
[98]	Two speech corpora: 1. The WSJ0 + 1 2. Italian ChildIt	171 children (86 male and 85 female) aged between 7 and 13 years	MFCC	HMM	SAT CMILSN MLLR	—	-Results demonstrated that the suggested normalising strategy outperformed MLLR adaptation under matched training and testing conditions -Speaker normalization is superior when training under unmatched acoustic conditions
[74]	PF-STAR database	198 speakers 4–8 years	MFCC	GMM-HMM	MAP MLLR	25.7%	-All age groups were significantly less affected by word errors with MLLR than with VTLN. -Models based on phonemes performed better, even with small vocabulary sets
[100]	1.Italian, ITC-irst Children's Speech Corpus	87 children aged 7 through 13 years	PMVDR	HMM	SMAPLR SAT	phonetic error rate of 26.5% for the 40 phonetic unit system and 20.8% for the reduced 33 unit system	-Colorado Literacy Tutor successfully incorporated the developed acoustic models -The authors translated the SONIC LVSR system from English to Italian to optimize the ASR

Table 7 (continued)

Reference	Language/ Corpus/ Data Set	Children Age Group (in years)/ Grade	Feature Extrac- tion Technique	Acoustic Modelling	Model Adaptation Algorithm	Accuracy/ WER	Remarks
[107]	Italian ChildIt corpora Italian SpontIt corpora	129 7 years to 13 years 21 8 years to 12 years	MFCC	HMM	SAT CMLSN	8.2% WER for adults and 10.2% WER for children	Speaker adaptive acoustic modelling ensures that the performance aligns with that achieved with group-specific acoustic models for adult and children -The speaking rate of children was analysed through time scale modification, to improve their ability to recognize speech - Combining speaking-rate adaptation with VTLN and explicit pitch scaling leads to additional improvements
[108]	British English, PF-STAR corpus	122 speakers 3–14 Years	MFCC	Context-dependent hidden Markov models (HMM), DNN-HMM	Speaking Rate adaptation	10.03% WER for adults + children Speech	Acoustic adaptation and pronunciation modeling were used to improve speech recog- nition for children
[109]	Children's Interac- tive Multimedia Project (CHIMP), The CU Read, Prompted Speech Corpus, CU Story Corpus CID	6 years to 14 years 6 years to 11 years 3rd through 5th grade (5–18 years)	MFCC, PLP and filter bank features	HMM	MLLT, SAT based on CMLLR	Performance is increased by 4.72% absolute with MLLT, whereas the WER is reduced by 11.94% absolute with SAT	

Table 7 (continued)

Reference	Language/ Corpus/ Data Set	Children Age Group (in years)/ Grade	Feature Extrac- tion Technique	Acoustic Modelling	Model Adaptation Algorithm	Accuracy/ WER	Remarks
[102]	British English PFSTAR	122 speakers 3–14 Years	MFCC	GMM-HMM	acoustic model interpolation and low rank features projec- tion	Relative improve- ment of 11.4%	To perform adaptation, the authors combined acoustic model interpolation and low- rank feature projections which reduces latency
[97]	British English PFSTAR	122 speakers 3–14 Years	MFCC	GMM-HMM	EV,SW- PCA+EV, SW- HLDA +EV	28% WER with EV, 19.83% WER with SW-PCA +EV, and 18.24% with SW- HLDA +EV	Low-latency adaptation approach is used to reduce latency through novel exploi- tation of the estimated VTLN warp factor
[110]	British English PFSTAR	122 speakers 3–14 Years	MFCC	DNN-HMM, GMM-HMM	Speaking rate adaptation (with variable frame- overlap and frame length)	RI up to 16%	In the context of mismatch ASR, the role of speaking rate adaptation was explored on DNN-HMM. Further, SRA combined with pitch modification is very effective for improving ASR
[111]	British English PFSTAR	122 speakers 3–14 Years	MFCC	DNN-HMM, GMM-HMM	Speaking rate adaptation (SRA) through variable frame- length and frame overlap	relative improvement of 16%	Structured low rank feature pro- jection was used to improve children's mismatched ASR
[103]	MyST Corpus: American English PFSTAR dataset: British English CMU Kids dataset	PFSTAR 122 3–14 Years	E2E	based on an encoder- decoder trans- former	Multilingual Whisper model & fine-tune wav- 2vec2	Both models had WER range of 14–16%	The whisper method may be better suited to unseen datasets, while wav2vec2 may be more suitable for real-time, task-specific applications

Table 7 (continued)

Reference	Language/ Corpus/ Data Set	Children Age Group (in years)/ Grade	Feature Extra- tion Technique	Acoustic Modelling	Model Adaptation Algorithm	Accuracy/ WER	Remarks
[104]	British English PFSTAR	122 speakers 3–14 Years	E2E	Transfer Learning	a Transformer pre- trained model	WER of 4.86%	To produce robust ASR for children's speech, authors used the source-filter warping data augmentation strategy on a transformer model pre-trained with adult speech -The proposed strategy outperformed on PF-STAR
[105]	My science tutor (MyST) children speech corpus: English OGI kids speech corpus	737 speakers 1100 speakers kindergarten to 10th grade	MFCC	Hybrid TDNN-F HMM, End-to-end acoustic model	CTC E2E system	—	An empirical study was conducted on End-2-End neural systems for children ASR
[106]	British English PFSTAR	158 speakers 4 to 14 years	MFCC, CNN- based end-to- end acoustic modeling	DNN-HMM, GMM- HMM, CNN-based end-to- end model	CNN-based end- to-end model	11.99% WER	CNN-based end-to-end acoustic modeling outperforms than those with the standard features like MFCCs

then with hybrid models. Comparative assessments are essential for understanding the evolution of E2E models.

Traditional Speech Recognition models and End-to-End models (E2E) differ in their architectural approaches to understanding spoken language and processing. Traditional speech recognition networks follow a modular approach, dividing the task into distinct stages with customized functions. End-to-end models adopt a more integrated approach, employing deep learning to directly learn representations from raw audio, reducing manual feature engineering. Many factors play into the choice between these approaches, including the availability of data and computational resources and the specific requirements for the ASR task. Traditional models employ pipeline architectures with separate components for acoustic modeling, pronunciation modeling, and language modeling [112–114]. MFCC features can be customized and applied to extract useful information from raw audio signals. The system uses these features as inputs. To generate the final recognized speech, the system decodes the input features using HMMs or other techniques. Conventional speech recognition typically requires the use of very large language models to achieve sufficient accuracy. In some cases, this can reach as large as 100 GB [112]. Speech recognition systems that use traditional methods include Hidden Markov Models (HMMs), Gaussian Mixture Models (GMMs), and Dynamic Time Warping (DTW). An E2E model simplifies the entire speech recognition process into a single, unified architecture. There are no intermediate representations between input audio and output transcriptions in these techniques. An end-to-end model uses deep neural networks in order to learn hierarchical representations from raw audio signals, eliminating the need for hand-crafted features. In these models, a single objective function is optimized to minimize the difference between predicted and target transcriptions over time. While end-to-end models simplify the architecture, they may require larger amounts of data and computational resources for training. A fully neural network-based E2E system requires much less memory than a conventional algorithm. Recurrent Neural Network Transducer (RNN-T), Attention-based Encoder-Decoder Models (AED), Monotonic Chunk-wise Attention (MoChA), Connectionist Temporal Classification (CTC), transformer-based speech recognition systems are examples of E2E speech recognition systems. It has been widely accepted that the HMM is the predominant framework for speech recognition, paired with the Gaussian Mixture Model (GMM). In recent years, however, many advanced models like the hybrid HMM-Deep Neural Network (DNN) model have begun to outperform HMM-GMM models [115]. A hybrid speech recognition model takes advantage of the strengths of both traditional and neural network-based approaches. Hybrid models have two main components: acoustic modelling and language modelling [114]. When it comes to acoustic modelling, hybrid models often employ neural networks for learning complex patterns and representations directly from raw audio signals. Neural networks are combined with HMMs to capture speech signals' temporal dependencies. Neural networks produce probabilities that HMMs use to model state transitions and generate final recognition [113]. In language modelling, hybrid models can use neural networks to capture long-range dependencies and context more efficiently. To handle language-specific constraints effectively, some hybrid models still use traditional language modelling techniques such as N-grams or Finite State Transducers (FSTs). Multiple hidden layers trained using novel techniques have shown superior performance over GMMs across multiple speech recognition benchmark tests, sometimes exhibiting substantial performance advantages. In comparison with fully neural network-based models, hybrid models can achieve superior performance with less training data [116]. It is possible to enhance the system's robustness in various acoustic conditions by combining neural networks with traditional models. The Deep Neural Network- Hidden Markov Model (DNN-HMM) and

Long Short-Term Memory- Hidden Markov Model (LSTM-HMM) are examples of hybrid models [113, 116].

Over the last decade, DNN-based hybrid modeling has resulted in a significant increase in the accuracy of automatic speech recognition (ASR). E2E models have several significant advantages over hybrid models. First, by directly outputting characters or even words, E2E models simplify the ASR pipeline significantly [114]. The design of hybrid models, however, requires a wealth of expertise and years of ASR experience. Second, E2E models utilize a unified objective function aligned with the ASR goal, optimizing the entire network. The traditional hybrid model, on the other hand, optimizes individual components separately and does not assure global optimum. Thirdly, E2E models are generally more compact than hybrid models since a single network is utilized for ASR. As a result, E2E models can be deployed with high accuracy on devices [113]. Finally, hybrid ASR systems are based on a distinct Language Model (LM). Comparatively, End-to-End (E2E) systems utilize a unified ASR model integrated with a network component serving as the Language Model. E2E systems are used to train the entire network [117]. Speech recognition models based on E2E represent state-of-the-art technology. Despite this, they are not yet able to surpass hybrid technology completely. For ASR advancement, E2E technology faces some challenges that require careful attention.

- i) To effectively learn diverse speech patterns, E2E models require a large amount of training data, which makes data collection even more critical. The available corpus for speech translation is notably smaller than the abundant data available for Automatic Speech Recognition (ASR) training [113].
- ii) Enhancing adaptation capabilities remains an essential focus for advancing the industry shift from hybrid to end-to-end (E2E) models. This emphasis is driven by the substantial accuracy improvements observed through adaptation, particularly for new speakers and domains.
- iii) The next significant challenge in the development of E2E speech translation systems is word reordering.
- iv) It is straightforward to incorporate new words that were not encountered during training into hybrid models. However, End-to-End (E2E) models find it difficult to incorporate such novel words.
- v) E2E models might be difficult to interpret, preventing the diagnosis and solving of potential problems because it is difficult to understand internal representations and decision-making processes.
- vi) In the end, training proficient End-to-End (E2E) models is consistently challenging for languages with low resources [113].

5.2 Linguistic mismatch

Language models are statistical models that assign probabilities to word sequences. By analyzing the previous words in a sentence, it helps predict whether a particular word or sequence of words is likely to appear. Language models are essential to any ASR system. It is possible to customize language models based on the application or task for which they are intended. It is very important to customize language models very carefully for children's ASR systems since they have a large number of variations in their corpus. There are several possible solutions presented for language models in the literature, just as there are for acoustic models. Two primary approaches to language modeling exist: the statistical

approach, which focuses on a relatively small corpus set, and the data-driven approach, which considers a much larger set of data. One language model can fit into both categories, as these concepts are not mutually exclusive. Formerly, language models were simply referred to as language models. However, with advancements in technology, pre-trained language models have come into focus, and older language models are known as conventional language models. It is necessary to explore the essential characteristics of these models.

- i) *Conventional language models (CLMs):* The purpose of these is to predict the likelihood of linguistic sequences in a causal manner. Known as auto-regressive language models, CLMs predict causal sequences. It is common for these models to calculate the joint probability by breaking down the text sequence into smaller units. In the pre-training era, CLMs were typically built from scratch using a training corpus, then used to determine text sequence probability for specific applications. CLMs assign elevated probabilities to natural text sequences that commonly occur in the real world. Consequently, they play a significant role in various applications such as text generation, speech recognition [118, 119], and machine translation [118, 120]. Prominent models in this category consist of N-gram language models [118, 121] exponential language models [118, 122], and preceding neural language models [118, 123]
- ii) *Pre-trained language models (PLMs):* Current approaches to language modeling focus on data-driven approaches, building neural network models based on extensive corpora. These models are called PLMs. PLM is fine-tuned with task-specific datasets and objectives customized for downstream applications. The PLM is versatile and pre-trained for various linguistic tasks and objectives, unlike CLMs. High-performance PLMs serve as the foundational components of numerous Natural Language Processing (NLP) systems. These models are beyond the limited functionality of CLMs and offer various types of language models. As deep learning has evolved, PLM training and usage have changed substantially compared to traditional statistical language models. Before being deployed for real-world tasks, PLMs undergo initial pre-training on extensive corpora, enabling them to learn universal representations encompassing both syntactic and semantic information.

Following pre-training, fine-tuning for specific tasks helps transfer acquired knowledge to different applications. However, the increasing capabilities of recent PLMs come at the cost of larger model sizes, higher training expenses, and heightened demand for training data, necessitating substantial computing resources and energy use. The practical applicability of these methods is limited by these factors.

Language models can be classified into various types:

- a) *Structured language models (SLMs)* predict linguistic units based on predefined language structures, such as dependency or constituent parse trees, instead of sequentially or reverse-sequentially. Leveraging these structures, SLMs strategically use language context to bring contextually relevant information closer to the unit they aim to predict.
- b) *Bidirectional LM:* Bidirectional language models (BiLMs) incorporate information from both directions rather than relying solely on causal contexts for predictions. Masked language models (MLM) are examples of bidirectional LMs. Bidirectional LMs aim to learn the inherent dependencies between linguistic units without supervision. Large-scale unlabelled corpora are used to generate semantic meanings for these models. In

contrast to CLMs focused on generating text sequences, pre-trained bidirectional LMs serve as a foundational framework, transferring learned knowledge through subsequent fine-tuning in diverse downstream applications.

- c) **Permutation LM:** Permutation language models combine the advantages of both CLMs and masked language models. Permutation LMs generate various permutations of input linguistic units by randomly shuffling their order as they receive an input sequence of linguistic units.

These models include N-grams, maximum entropy models, and neural networks as popular approaches to modeling probability distributions of text sequences.

The following architectural types can be used to categorize language models:

- i) **N-gram:** An N-gram is a sequence of N consecutive tokens. N-gram language models operate under the assumption that a token's probability is solely influenced by its preceding N-1 tokens and remains independent of other contextual factors [121, 124]. N-grams are computationally efficient and simple. They can also be used to model short-range dependencies.
- ii) **Maximum Entropy Models:** Maximal Entropy models, also known as exponential models, use feature functions to calculate the probability of text sequences. These features are derived from N-grams [122].
- iii) **Feed-forward Neural Network (FNN) Models:** These models use historical contexts as input and produce token probability distributions as outputs. Using a fixed window, this model gathers contexts of fixed length, serving as a neural counterpart to N-gram language models. Its advantages include the ability to handle unseen N-grams by representing each token as an N-gram within a dense vector space. A further advantage of this model is that it does not need to count and store transition probabilities, which is a characteristic of traditional N-gram models. Some problems of FNN are i) Lack of Sequential Context. ii) FNNs cannot model complex linguistic structures because of the lack of natural sequential dependencies between words.
- iv) **Recurrent Neural Network (RNN) Language Models:** These models include recurrent neural networks to model sequential dependency between words, allowing the capture of context over time. As RNNLMs use the entire preceding history to predict the next token, they do not require the Markov assumption. However, the inherent challenge of gradient vanishing in RNNs prevents the effective learning of the model [125]. Long-range dependencies on sequential data can be captured by RNN LMs. These are suitable for tasks where context over time is crucial. It is difficult to train RNN LMs quickly, due to the lengthy training process and high computational load. The vanishing gradient problem also affects these models.
- v) **Long Short-Term Memory (LSTM) Language Models:** These are RNNs with memory cells, which allow language models to capture long-term dependencies. A LSTM addresses the issue of vanishing gradients in RNNs and is effective at modeling long-term dependencies. LSTM LM remains complex due to its high computational complexity.
- vi) **Transformers:** Transformer architectures capture long-term dependencies and crucial sequence components by utilizing a self-attention mechanism. During training and inference, transformers are highly parallelizable. The architecture consists of an encoder and a decoder. Some benefits of transformers are: i) they are highly parallelizable, which means they can be trained more quickly. ii) Efficient at capturing global context with

self-attention mechanisms. A transformer also has weaknesses i) It requires substantial computing resources and it can also struggle with very long sequences due to positional encoding.

Hybrid ASR systems employ a distinct language model during the inference stage. E2E systems, on the other hand, use a single ASR model with a network component serving as a language model. E2E systems train the entire network together. In RNN-T, the prediction network, and in AED models, the decoder performs the function of an LM. Consequently, E2E ASR does not require distinct and separate LMs [117]. There are multiple methods for integrating language models into E2E ASR. However, many studies have shown that separate LMs help improve E2E ASR recognition accuracy. Different approaches to LM fusion have been explored to enhance the accuracy of ASR under various circumstances. These methods are [117]

- i) **Shallow fusion:** This is the most commonly used method for combining E2E and language models at the time of inference.
- ii) **Deep fusion:** In this approach, language models and E2E models are integrated through a joint network. The joint network is trained with all parameters considered collectively, promoting improved collaboration between the models to enhance recognition accuracy [126].
- iii) **Cold fusion:** This approach combines the LM and the E2E model similar to Deep Fusion. During cold fusion, the E2E model is trained with a pre-trained LM, while the LM parameters remain fixed or "frozen" [127].
- iv) **Internal LM estimation:** a shallow fusion strategy is another way to mitigate language bias in training data. The problem of language bias arises when the source domain (training data) does not match the target domain (test data).

Transformers with Bidirectional Encoder Representations (BERT) [128] and Generative Pre-trained Transformer GPT-2 [129] are exemplary models based on Transformer language models. These language models derived from the Transformer architecture have found numerous applications in end-to-end ASR systems. Some notable implementations include N-best rescoring [130] and the incorporation of dialog context embedding [131]. Many studies have indicated that adding separate language models enhances E2E ASR accuracy. There are three potential reasons why E2E ASR may still require an external component.

- i) **Compensation for poor generalization:** As compared to traditional modular-based models like acoustics, E2E models include a more sophisticated mapping function. Due to this, E2E models are prone to overfitting issues, especially when training data volume is inadequate. Pre-trained LMs can potentially address this challenge by compensating for less general predictions made by E2E models.
- ii) **Use of external text data:** The E2E model requires both paired speech and text data, while the LM model only requires text data for training. Text data collection is generally easier than obtaining speech data. Furthermore, for equivalent numbers of sentences, LMs provide faster training speeds than E2E models. Due to this, the LM can be enhanced more efficiently by utilizing external text data, thereby offering supplementary performance improvements to the ASR system. LM performance can be improved more effectively with external text data, increasing ASR performance.

- iii) **Domain adaptation:** When applying an E2E model to a particular domain, domain adaptation enhances recognition accuracy. The E2E model requires sufficient paired data from the target domain to be adapted to that domain. It may also be very costly to manage multiple E2E models for various supported domains. If a pre-trained language model for the target domain is available, it is more likely to improve recognition accuracy for domain-specific vocabulary and speaking styles without having to update E2E.

Additionally, pronunciation and language modelling are issues that need to be addressed in addition to creating customized acoustic models for children. It may be unsuitable for children with poor pronunciation or for children younger than 7 years old to use an ASR system with a standard pronunciation dictionary. Several works have stressed the importance of customized language models in recognizing children's speech [5, 19, 42, 53, 70]. According to the article [19], a language model for creating an Automatic Reading Tutor (ART) system can be efficiently and robustly built. The approach is based on Microsoft Research's proposed interpolated LM approach. This language model offers four advantages. The first is that it reduces the effort required to build a new ART system, which is new and challenging due to the target users of the system being children. In addition, it provides robustness in detecting different types of reading mistakes without having to predict which one's children would make. As a third advantage, by modifying the weights of the garbage model, a ROC curve can be obtained instead of a fixed point. Finally, it is very efficient and does not have much overhead. It is also necessary to improve language model's ability to detect more reading mistakes in the future, along with more training data and better training approaches for improving the children's acoustic model. Instead of using a word-level garbage model, which might be more robust to subword level mistakes, a phoneme-level garbage model might be more appropriate. Research has demonstrated that using a children's speech-trained language model contributes significantly to their recognition ability [16, 70, 132]. Data sparsity was compensated for by reducing the tied-states of a speech recognizer [132]. There is considerable evidence that pronunciation variation affects children's speech recognition performance [10]. It has been shown in [57] that using a user-customized pronunciation dictionary can improve pronunciation performance. Children with good speaking skills perform similarly to adults, but their error rates are at least four times higher than those with poor pronunciation. It was also found in the result that the performance of children's ASR increases if the customized dictionary is used, but this improvement is very less.

6 Data augmentation

The demand of children ASR system has increased from last few years due to the numerous applications of ASR specifically for children. For building a successful ASR system large amount of speech data is required. However, data collection from children is very challenging and time consuming task [27]. Therefore, data augmentation is efficient alternative to deal with the issue of data scarcity.

It is a frequent approach for increasing the amount of training data, avoiding overfitting, and improving model stability [133]. By decreasing the influence of acoustic variability in children's speech across speech recognition toolkits, data augmentation enhances speech recognition performance for children in general. A variety of data augmentation techniques can be used to enhance the dataset, including pitch change, speech change,

tempo change, volume change, reverberation change, and spectral change [134]. On the other hand, vocal tract length perturbation (VTLP), an object recognition method using neural networks, relies on adding new datasets without modifying their labels. In order to overcome the problem of data scarcity, some of the commonly used schemes for data augmentation are semi-supervised training [133, 135, 136], multi-lingual processing [136, 137], acoustic data perturbation [138] and speech synthesis [75] that can be applied on original data to generate synthetic data. Jaitly & Hinton [138], in 2013, proposed augmenting the training database by randomly generating a random warp factor for each utterance during training. By averaging multiple predictions over the same set of data, improved decoding results can be achieved. A random, linearly warped variant of spectral data can significantly improve recognition accuracy since the data can be generated at random and linearly warped. Speed perturbation for audio has low implementation cost and it was reproduced by the combination of tempo perturbation and VTLP [133]. The obtained results show improvement in WER than either of above mentioned methods.

Park and coworkers [139] in 2019 applied SpecAugment, another data augmentation scheme, directly to a neural network's data inputs (i.e., filter bank coefficients). Warping characteristics, frequency channel masking blocks, and time step masking blocks are all part of the augmentation strategy that were explored in SpecAugment. The following were the main findings from the experiments: Augmentation turns an over fitting problem into an under fitting problem. Under-fitting is a common problem that may be solved in a variety of ways. Time warping helps, but it isn't the most important component in enhancing performance. Label smoothing might make training more insecure. Therefore, the authors concluded that the performance of ASR networks is considerably improved by applying SpecAugment.

6.1 Data augmentation for child ASR

Using data augmentation to deal with data scarcity has been found to be effective for improving the performance of DNN acoustic models for normal speech recognition [140].

Chen and coworkers [134] in 2020 worked on the "Ethiopian" method which is presented at the SLT 2021 (Spoken Language Training Workshop) Children Speech Recognition Challenge. To address children's speech issues, a variety of data processing and augmentation strategies have been offered. The acoustic variability produced by children's shorter voice tracts was suppressed using VTLN. The tests were carried out using a standard chain model, in which the alignments were obtained by training a GMM-HMM model. Data augmentation has been demonstrated to increase children's speech recognition skills in Kaldi. Data augmentation improves the model's robustness to acoustic fluctuation in a child's speech.

There have been many attempts to alleviate a lack of data by using simple data augmentation setups involving adding un-modified adult speech data to children's training set [141]. However, this approach has generally not been successful [50, 142]. Therefore, in-domain augmentation techniques such as VTLP and Stochastic Feature Mapping (SFM) were explored by various researchers, amongst which SFM gain particular interest, since SFM technique does not depend upon hyperparameters. Fainberg and co-authors [141] in 2016 applied SFM technique on out-of-domain data and concluded that when SFM is used, out-of-domain SFM showed 6.2% relative improvement in WER compared with that of DNN baseline.

Shahnawazuddin and co-authors [143], in 2017, proposed a novel approach by adding two modules to SMAC feature extraction to improve the noise and pitch robustness. The resultant approach is named robust SMAC (R-SMAC). They also explored SMAC and MFCC features on noise and found that the noise robustness of SMAC features is greater than that of conventional features such as MFCC. SMAC features, however, appear to be sensitive to pitch variations as well. It was proposed that pitch sensitivity could be reduced using spectral smoothing and adaptive filtering. Research conducted by the researcher employed MFCC, SMAC, and RSMAC feature extraction techniques. Additionally, the R-SMAC features demonstrated superior pitch robustness while maintaining noise immunity.

It is possible to change the acoustic attributes of adult speech samples to resemble that of children by using cycle consistent generative adversarial network (GAN). Shahnawazuddin and his team [144], in 2020, worked on VC-based out-of-domain data augmentation, which significantly reduces WERs. In order to prevent acoustic mismatches caused by differing speaking rates, cycle-GAN-based voice conversion is explicitly time scaled in children's speech tests. Using the speaking rate adaptation method with data augmentation, further improvements can be made. Combining VC-based data augmentation with time-scaling of test utterances results in a 21.1% improvement.

Kadyan and co-workers [145], in 2022, proposed a feature warping method. The authors explored a fused feature extraction approach based on Mel frequency-gamma tone frequency cepstral coefficient (MF-GFCC). Further, in-domain data augmentation is used to overcome the issue of data scarcity. For this purpose, artificial data is generated by combining noisy and clean corpora. Various hybrid/fused algorithms, including MFCCPLP and GFCC-PLP, have been applied for more robustness in almost constant real-life auditory systems. A combination of original and augmented datasets was used to experiment with the efficiency of the whole system. Initially, the experiments were carried out using real datasets. The training of hybrid acoustic classifiers is then used to improve efficiency over training from only existing speech corpora. The results have shown that the MF-GFCC feature vectors improved by 20.59% in a noisy environment and by 19.39% in a clean environment. Further, for limited data scenarios [28], in-domain data augmentation employs original signal and noise-induced synthetic data to supplement the data. Pitch features were extracted from Punjabi, and the test set was normalized using a VTLN approach to accommodate the tonal characteristics of the language. The training model has been enhanced artificially with multiple noise combinations viz. Babble, Factory, and Volvo at various SNR values. As compared to male-based data, the blended female adult-child speech reduced acoustic mismatch in training data. Various data augmentation techniques are summarized in Table 8.

7 Low resource languages

Throughout the past few years, speech recognition technology has made significant progress, facilitating smooth interactions between humans and machines. In the beginning, ASR development was focused on international languages, commonly British English and American English. Most of these advancements target major languages with abundant resources and data, overlooking the linguistic richness of low-resource languages.

Languages that suffer from a shortage of resources include transcribed speech data, pronunciation dictionaries, language knowledge, text annotation, and more. These languages are categorized as low-resource languages [147, 148]. Furthermore, these

Table 8 Data Augmentation Techniques

Reference	Language/Corpus/ Data Set	Children/ Age Group	Feature Extraction Technique	Acoustic Modelling	Data Augmentation Techniques	Accuracy/WER	Remarks
[134]	Mandarin Set C1 data- set Reading Speech Mandarin Set C2 dataset conversation speech	927, 7 to 11 years 54 4 to 11 years	—	HMM-GMM	Speed perturbation Volume perturbation Pitch perturbation Tempo perturbation Reverberation pertur- bation Spectral augmentation	21.66% CER on Track 1 of evaluation set 16.53% CER on Track 2 of evaluation set	Data augmentation for children's voice recognition has been shown to be easy and effective; however, the authors didn't use the data augmentation in the proper way
[141]	British English, PF- STAR	4 -14 Years	MFCC	HMM-GMM HMM-DNN	out-of-domain Stochastic Feature Mapping	6.2% relative improve- ment	The out-of-domain SFM provides better results than using in-domain SFM on PF-STAR
[146]	British English, PF- STAR	182 4 -14 Years	MFCC	TDNN	Out-domain augmenta- tion together with Cycle-GAN for voice conversion from adult data	21.1% relative improvement obtained with VC- based data augmen- tation	VC-based data augmen- tation with time-scal- ing of test utterances results in 21.1% RI
[86]	British English, PF- STAR	122 4 -14 Years	MFCC R-SMAC	DNN-HMM	Using white and bab- ble noises at 5 dB and 10 dB for noise- based augmentation	—	Pitch-robustness of the existing SMAC features could be enhanced using the proposed acoustic features along with the retained immunity toward additive noises

Table 8 (continued)

Reference	Language/Corpus/ Data Set	Children/ Age Group	Feature Extraction Technique	Acoustic Modelling	Data Augmentation Techniques	Accuracy/ WER	Remarks
[145]	Punjabi	—	hybrid MF-GFCC	HMM-GMM HMM-DNN	In domain noise-induced (using white noise, speech babble, and factory) data augmentation explored	20.59% and 19.39% relative improvement was observed for noisy environment and clean environment, respectively	A combination of clean and noisy corpora is used to generate artificial data for implementation
[28]	Punjabi	32 Speakers	GFCC + Pitch + VTLN	DNN-HMM	Artificially enhancing the training model with multiple noise combinations- Babble, Factory, and white at various SNR values	the relative improvement of 30.94%	Pitch features were extracted from Punjabi data set, and the test set was normalized using a VTLN approach to accommodate the tonal characteristics of the language

languages frequently encounter challenges such as the absence of comprehensive linguistic resources, the lack of a distinct writing system, limited linguistic expertise, scarcity of web resources, and inadequate electronic materials like speech corpora, lexicons, and language models crucial for training and developing robust speech recognition systems [147].

There are over 7,000 languages globally, and it is estimated that at least 40% of them are endangered, with approximately half lacking written forms [148, 149]. Languages are disappearing at an alarming rate in today's world, with estimates indicating that nearly 50% of current languages will disappear within the next century [147, 150]. The extinction of endangered languages is an irreparable loss to the world and is destroying the diversity of culture. A lot of attention is being paid to protecting endangered languages by international linguistic communities right now. These communities are collecting audio and text data of these languages and establishing a digital database of them. However, the task of processing speech corpora is extremely difficult. To enhance the comprehensibility of these corpora, all audio and video recordings acquired through field investigations require manual annotation. Due to the scarcity of native speakers or professionals available to perform this manual annotation process, the manual annotation process requires a considerable amount of human effort. Due to this, a large number of original audio and video materials in multiple languages accumulate without being adequately processed. Therefore, the first and most important reason for developing ASR in under-resource language is to preserve literature and best practices for coming generations. It is one of the best methods to save the language's cultural heritage. In addition, research in this area is encouraged by widespread applications of ASR, such as assistive tools for physically disabled people, railway stations, and customer service through voice-based interfaces [151]. In low-resource languages, conventional ASR systems may not be as effective as they should be. The field of study is associated with low-resource speech recognition, emerging as a prominent and crucial challenge in the area of ASR [148]. It is challenging to develop ASR for low-resource languages. Following are some challenges associated with low-resource languages.

- i) A significant obstacle lies in the challenge of finding native speakers who possess the required technical skills for developing ASR systems in their native languages. Bridging the gap necessitates collaboration between technology experts (system developers) and language experts (native speakers) [147].
- ii) the availability of high-quality speech corpora or the unavailability of resources and corpora [152].
- iii) low-resource languages often suffer from a lack of well-addressed linguistic literature, numerous dialects, and intricate phonetic nuances, representing inherent instabilities.
- iv) there is issue of pronunciation variability, where low-resource languages may lack standardized pronunciation guidelines [147].
- v) a significant challenge persists in low-resource languages, particularly those with limited audio and text data availability. This complexity is increased by the existence of multiple dialects, as observed in Indian languages.

Due to advancement of research in speech recognition, researchers' attention shifted to implementing ASR in the national languages. Further, due to ASR's widespread application, research has shifted to developing ASR in regional languages. Interacting with a machine in their native language generally makes people feel more comfortable [153].

In India, ASR is in its early stages of development in the aspect of recognition of utterances spoken by the various users in different languages of their region. Only the Hindi

language has commercially accessible ASR devices till now and few Indian languages, including Hindi, Marathi, Malayalam, Tamil, Telugu, and Bengali, have well-developed ASRs, while others are still under development. In comparison to non-Indian languages, Indian language voice recognition research has not yet reached that level of perfection. Speech recognition research for local Indian languages is still underway. Due to the existence of dialectal and tone differences within the territory of multiple languages, substantial challenges arose. It raises the challenge of assembling a large corpus in a specific language. It's challenging to create a speech corpus for an Indian language. Low-resource languages face significant challenges, particularly those with the limited availability of both audio and text data. This complexity is increased by the presence of multiple dialects, as observed in Indian languages. Despite this, many Indian languages can be categorized into the same language families, sharing common scripts and grammatical structures. In such cases, adaptation and fine-tuning techniques become crucial for overcoming the inherent low-resource nature of the data by leveraging well-resourced languages that exhibit similarities [154].

Recent developments in multilingual ASR have made it possible for a single system to accommodate multiple low-resource languages by utilizing limited labeled corpora across multiple languages. As multilingualism becomes increasingly prevalent, code-switching ASR is also becoming more popular. Code-switching involves the free interchange of multiple languages within a single sentence or between sentences.

To develop an effective multilingual and code-switching ASR, diverse languages' acoustics, linguistic characteristics, and available data need to be considered carefully [155].

Due to a scarcity of code-switching corpora, identifying code-switching patterns across various languages and creating code-switching automatic speech recognition systems remains challenging.

A few examples of LRLs at the national and international levels will be discussed in the next paragraph:

In [156] the researchers worked on Seneca a low-resource endangered language spoken in North America. In this work, the authors show that word error rates are up to 40% lower than those reported using traditional GMM-HMM methods and well-established deep neural techniques. As a result of combining a deep acoustic model with a trigram language model, better performance is achieved than with a traditional GMM/HMM model without the need for transfer learning or data augmentation. The authors of [157] also worked on Seneca language. In [158] the authors worked to develop an ASR system for Sinhala. It focuses on the Sinhala language, one of Sri Lanka's official and national languages, which are widely spoken. To enhance the acoustic modeling process, experiments were conducted with a variety of deep neural architectures, including pre-trained DNN, DNN, TDNN, and TDNN + LSTM. A GMM-HMM statistical model is also trained on the same dataset, serving as the baseline model to compare the effectiveness of the deep learning approach. Results indicate that deep neural network architectures outperform statistical baseline models. This produces a lower Word Error Rate (WER) of 7.48% on the test dataset. Researchers conducted experiments on the Babel Turkish dataset for low-resource ASR in [159]. An investigation has been conducted on the performance of neural network architectures when there is a limited amount of available data. Employing multi-layer bidirectional LSTM networks, a fully convolutional architecture inspired by Wav2Letter, and exploring the DeepSpeech-like architecture developed by Salesforce (DS-SF). The experiment setup introduced a promising CTC-loss modification that incorporates segmentation during training. Using in-domain data alone, the final system achieved a 45.8% WER, the best result ever published for

Turkish end-to-end systems. A well-tuned end-to-end system achieves results that are very similar to hybrid DNN-HMM systems, even for languages with limited resources.

Low-resource multilingual experiments can benefit from Connectionist Temporal Classification (CTC). Compared to hidden Markov models, CTC-based systems are generally easier to train [160]. In this study, authors utilize several languages from IARPA's Babel project to evaluate the model. The languages included in this study are Turkish, Haitian, Kazakh, Mongolian, Amharic, Tamil, Tagalog, Pashto, Kurmanji, and Swahili. Further, they demonstrate the feasibility of training multi-lingual and cross-lingual acoustic models directly on phoneme sequences, eliminating frame-level state labels. In multi-lingual settings, training on related languages or large amounts of clean data appears advantageous, while there is no apparent benefit from training on numerous languages alone. As a result, Switchboard and Babel are capable of combining data from different sources. The speech-to-text system in the Slovak language was presented in [161]. Transcribing broadcast news and conversations was the purpose of this system. The Slovak language belongs to the west Slavic family of European languages, and approximately 7 million people mostly speak it in Slovakia. In this system, authors used a large N-gram language model & LM is estimated using 63 million words and context-dependent acoustic models. Using a rule-based pronunciation model, a pronunciation dictionary is generated. The system incorporates multilayer perception (MLP) features imported from English, conceptual with cepstral PLP and F0 (pitch) features.

Some significant studies presented in [162] are a) training data for Iban, which is a low-resource language, is inadequate and there is no pronunciation dictionary available. The problem was addressed by using resources from a closely related Malay language. For the generation of a pronunciation dictionary semi-supervised method was used and cross-lingual strategies for developing the acoustic models that can train with minimal data. b) In another study a multi-speaker text-to-speech corpus was generated in Javanese and Sudanese (two low-resource languages of Indonesia). This text-to-speech application was built using a multi-speaker crowd-sourced corpus in Javanese and Sudanese. c) In another work, the Gujarati language (Indian language) was studied. In this ASR a deep learning-based E2E model was adopted. d) Furthermore, in another study of Assamese, an LRL of India, the ASR system was designed with traditional HMMs in association with GMMs, with MFCC as the characteristic of the model. Further, SGMM, deep neural networks, and HMM models were used in integration. The system underwent assessment using the threefold cross-validation method, and the best ASR system demonstrated an average vocabulary error rate of 4.3%.

On the international level, two projects focus on low-resource languages:

- A) Intelligence Advanced Research Projects Activity (IARPA) BABEL PROGRAM: The IARPA Babel program aims to make speech processing technology flexible and resilient, so it can adapt quickly to any language. In this project, analysts will be able to search and process extensive amounts of real-world recorded audio [148]. As part of the Babel program, data is gathered from over twenty-three low-resource languages, allowing multilingual experiments related to feature extraction and acoustic modeling to be conducted. These languages are Turkish, Tagalog, Pashto, Georgian, Cantonese, Bengali, Assamese, Zulu, Vietnamese, Tamil, Swahili, Lao, Kurmanji Kurdish, Haitian Creole, Tok Pisin, Telugu, Kazakh, Cebuano, Lithuanian, Igbo, Guarani, Amharic, Dholuo.
- B) DARPA Low Resource Languages for Emergent Incidents (LORELEI) PROGRAM: This program aims to improve the efficiency of human language technologies in low-

resource languages and ease the process of quickly retraining them for new languages. The primary focus is on deploying resources rapidly in emergent situations, particularly during sudden events like natural disasters [163].

Here are some international tools and apps for low-resource languages: *a) Voice search in three South African languages:* A prominently visible outcome, both in terms of visibility and commercial relevance, stemming from this initiative was the creation of applications capable of conducting web searches using spoken queries in three South African languages: isiZulu, South African English, and Afrikaans [147]. *b) Interactive voice forum for farmers in rural India:* Avaaj Otalo was started in Gujrati in the summer of 2008. Farmers in India, many of whom have limited formal education, can use a voice message forum. The platform was designed to offer interactive, on-demand access to agricultural knowledge. *c) The PI project:* The PI project, funded by the French ANR (Agence Nationale de la Recherche), was entirely focused on the development of ASR for under-resourced languages, with a specific emphasis on languages from Vietnam, Laos, and Cambodia. *d) The Rapid Language Adaptation toolkit (RLAT):* The SPICE project (NSF, 2004–2008), conducted at the Language Technologies Institute at Carnegie Mellon, and the Rapid Language Adaptation project at the Cognitive Systems Lab (CSL) had the objective of closing the gap between language and technology expertise [147].

The characteristics of LRL languages are discussed in this section. In brief low-resource languages refer to languages with limited linguistic resources, often in terms of available data, tools, and research. In the context of India, there are several languages that fall into the category of low-resource languages. India is known for its linguistic diversity, with many languages spoken across the country. While some languages have significant resources and documentation, others may face challenges in representation and support.

7.1 Review of literature in Punjabi speech language

The speech research community has focused a lot of attention on ASR for under-resourced languages over the last decade. One of the under-resource languages is the Punjabi Language, a very popular language in northern states of India, specifically Punjab. There are 113 million Punjabi speakers worldwide. It is the world's 9th most popular language. Punjabi is 11th most broadly spoken language in India and the Indian Subcontinent's 3rd most spoken language. Punjabi is England's 3rd most widely spoken native language and Canada's fifth most widely spoken native language. Although Punjabi is spoken by approximately 105 million people worldwide, still the resources are insufficient, which makes Punjabi-ASR a challenging endeavor. There is limited literature on adult speakers of the Punjabi language [164–168]. In addition, researchers have established their own corpus to conduct various research activities. Therefore, the lack of data and limited resources are important reasons for the unavailability of Punjabi-ASR.

A limited amount of adult speech data is freely available in the Punjabi language, and to the best of our best knowledge, very few studies are reported in the case of children ASR. Data paucity and restricted resources are important factors for the non-availability of Punjabi ASRs. Due to abovementioned factors developing a child ASR in the Punjabi language becomes very challenging. Since there is a scarcity of child speech data in the Punjabi language, hence relatively very few studies are reported on children's speech [28, 145, 169–172]. This is not for the Punjabi language, but overall very less work has been carried out on child speech. To overcome the aforementioned problems, the ASR systems

are usually trained on adults' speech data and tested on children's speech data. However, degraded performance recognition has been reported for such ASR systems. Such types of data are referred to in the literature as '*mismatched data*.' The possible reason behind the degradation performance is variation in acoustic characteristics of speech with the type of speaker, emotions, age, gender, etc. Further, the average phoneme duration in the case of children is more compared with adults and hence significantly affects the speaking rate of children (lower in children), variation of which is much higher within the child speakers. Moreover, the geometry of the vocal organs of children and adults differ considerably (smaller in children), which results in the scaling of fundamental frequency or pitch [111].

To train ASR system, large volumes of data are required, however, standard Punjabi speech corpus is not freely available for research purposes. Further, limited available child corpus, on the other hand, is not widely accessible [5, 6, 15, 26]. This could be because of the various problems associated in collecting speech corpora, particularly in the case of the Punjabi language, which is considered as a low-resource language.

The work on Punjabi ASR was initiated by Kumar and Singh in 2011 [165] authors developed the system using HMM and dynamic time warping (DWT) techniques performed on the small vocabulary of isolated spoken words. Dua and co-workers in 2012 implemented an isolated Punjabi ASR system for adults. They achieved overall performance (open environment) was 94.08% accuracy with MFCC and HMM [164]. Other researchers who worked on adult speech are [145, 151, 167, 168]. Children's ASR has, however, received very little attention in the Punjabi language [28, 171–175]. Kaur & Kadyan [176] used bMMI (boosted maximum mutual information) discriminative technique to implement the children's ASR system in the Punjabi language. An improvement of 22–26% accuracy is shown in the results. In children's Punjabi ASR system, Bhardwaj and his team [171] in 2020 applied the Subspace Gaussian mixture model (SGMM) acoustic modeling technique. The experimental results showed an accuracy of 83.66%.

Taniya and co-workers 2020 [175] implemented the Punjabi ASR system on child speech. They used MFCC for feature extraction and DNN modeling techniques have been applied to four hours of Punjabi children's speech corpus. There was an 87% accuracy rate obtained. Most of the researchers used either MFCC for feature extraction [28, 171, 175] or MFCC fused with other feature extractions techniques like MFCC-FDLP hybrid approach [177], hybrid MF-GFCC [145]. Various researcher [28, 171, 172, 178] explored DNN-HMM acoustic modeling on child ASR in the Punjabi language. Further, GMM-HMM and DNN-HMM acoustic modelling were explored by bhardwaj et al. (2020) and [179]. In 2021, Bawa & Kadyan [28] investigated in-domain data augmentation on originally collected data and induced noise to obtain synthetic data. Pitch features also have been extracted to investigate Punjabi language tonal characteristics, and a VTLN approach has been used to normalize the test set. The results show that the authors achieved a relative improvement of 30.94%. Recently, Kadyan and coworkers in 2022 [145] generated synthetic dataset using factory, babble, and white noise to augment in-domain data. The authors reported 20.59% relative improvement in noisy environments and 19.39% in clean environments. Recently, A prosody-based out-of-domain augmentation method has been studied by Kadyan and co-authors in 2021 [178] to deal with the scarcity of training data. Adults' speech is modified specifically to resemble that of children by modifying its pitch and speaking rate. Several methods have been explored to reduce the acoustic mismatch, including vocal-tract length normalization (VTLN), explicit pitch, and duration modification. Three approaches have been explored, and all three are highly effective.

Table 9 represents the literature reported on the Punjabi ASR developed in the case of children.

8 Analytical summary and discussion on child ASR

This section presents an analytical discussion and summarizes the need, applications, acoustic and linguistic characteristics of children speech, children corpus, and various challenges of children's ASR system along with their possible solutions. Further, this work also highlights the comparative summarization of research work done by various researchers in different approaches of mismatch system in children's ASR, data augmentation, low resources languages, which includes features extraction techniques; acoustic models, and performance measures.

ASR system for children is as important as that of adults, since children are more dependent on these systems nowadays, such as computer games, reading tutors, foreign language learning tools, etc. Development of Children Speech Recognition System is becoming increasingly important because of the two main reasons; due to their widespread applications in education and assessment, entertainment, language learning tools, and medicine as reported in detail in Section 2; and the use of a speech interface is beneficial for children who cannot operate traditional input devices such as a keyboard, mouse, and pointer.

However, it takes more time and efforts to design an ASR system for children's speech than for adults' speech because of *variety of challenges* that exist for the development of children's ASR, such as: the collection of children's speech data is a very complex task; there is no publicly available child corpus particularly in low resource languages; children speakers differ greatly due to acoustic & linguistic variations, and ASRs developed for one age group are not suitable for another age group.

Speech characteristics of children are broadly classified into two categories; i) Acoustic characteristics: Children's acoustic characteristics change rapidly with age and are also widely different from adults. The anatomical and physiological changes that occur throughout with child's growth are the prime reason for rapid changes in acoustic characteristics ii) Linguistic characteristics: Linguistic mismatch in children's speech plays a vital role along with acoustic mismatch. Since a child has not fully developed the knowledge of a language, he/she cannot use the rules of that language, which leads to linguistic variabilities and affects ASR's performance.

There are *two approaches to develop an ASR* system for children. One approach is to collect children's speech data separately and build the whole system using only children's speech. Such type of ASR is called *matched* ASR. However, due to a dearth of publicly available children's speech data, constructing a well-trained ASR framework is challenging. The second approach is to build the system using adult speech data means when ASR is trained on adults' speech data and evaluated/tested on children's speech data, or vice versa. Such type of ASR called *mismatched* ASRs.

Acoustic mismatch and variability have been recognized as the primary causes of performance deterioration in the development of high-performance ASR applications for youngsters. In order to improve acoustic differences between adults' and children's speech, different methods have been explored in literature which may be clubbed into *three distinct groups* based on the field in which different acoustic sources of mismatch are addressed: feature domain; signal domain, and model domain.

Numerous researchers worked on the domain of features to reduce the mismatch between child speech and adult speech and found significant improvement in the performance of child ASR systems. The overview of available research on feature domain is shown in Table 5. To minimize noise components in input noisy signals, the signal domain method uses single-channel speech improvement algorithms as a preprocessor for

ASR. The design of signal models has traditionally been influenced by the robustness of the models versus background acoustic noise. Speech recognition technologies today are highly sophisticated as the ASRs themselves contribute to noise robustness. The overview of available research on signal domain is shown in Table 6. A model domain approach is a powerful tool for reducing the mismatch between statistically trained recognition models by means of adaptation techniques such as MLLT (maximum likelihood linear transform), MAP (maximum a posteriori), etc. ASR systems have to adjust their models in order to compensate for mismatches between their test and training data. However, model adaptation is a very challenging and complex task because in it large, complex models need to modify without explicit supervision using a limited amount of target data. The overview of available research on model domain is shown in Table 7.

A methodical comparison is conducted across different domains, specifically the feature domain, signal domain, and model domain, to address acoustic mismatch. The referenced studies exhibit a spectrum of approaches, with some researchers opting for conventional architectures like Hidden Markov Models (HMM), others employing hybrid techniques such as Deep Neural Network-Hidden Markov Model (DNN-HMM), Gaussian Mixture Model-Hidden Markov Model (GMM-HMM), and some acknowledging contemporary End-to-End (E2E) such as Recurrent Neural Network Transducer (RNN-T), Attention-based Encoder-Decoder models (AED), Monotonic Chunk-wise Attention (MoChA), Connectionist Temporal Classification (CTC), transformer based speech recognition systems. This diversity in methodologies underscores the multifaceted nature of addressing acoustic mismatch in the field.

Additionally, pronunciation and language modelling are issues that need to be addressed in addition to creating customized acoustic models for children. It may be unsuitable for children with poor pronunciation or for children younger than 7 years old to use an ASR system with a standard pronunciation dictionary. The choice of specific models, architectures, and parameters will depend on the specifics of dataset and the characteristics of children's speech aim to recognize. It may also be beneficial to explore state-of-the-art ASR models and architectures designed for children's speech recognition.

The demand of children ASR system has increased from last few years due to the numerous applications of ASR specifically for children. For building a successful ASR system large amount of speech data is required. However, data collection from children is very challenging and time consuming task. Therefore, *data augmentation* is efficient alternative to deal with the issue of data scarcity. By decreasing the influence of acoustic variability in children's speech across speech recognition toolkits, data augmentation enhances speech recognition performance for children in general. A variety of data augmentation techniques can be used to enhance the dataset, including pitch change, speech change, tempo change, volume change, reverberation change, and spectral change. Table 8 shows the various data augmentation techniques that are used in Children's ASR system.

In recent years, there has been significant progress in speech recognition technology, facilitating smooth human-machine interaction. In the beginning, ASR development was focused on international languages, commonly British English and American English. Most of these advancements target major languages with abundant resources and data, overlooking the linguistic richness of low-resource languages. The languages that suffer from a shortage of resources, encompassing transcribed speech data, pronunciation dictionaries, language knowledge, text annotation, and more, categorizing them as low-resource or under resource languages. Therefore, the first and most important reason for developing ASR in under-resource language is to preserve literature and best practices for coming generations. In addition, research in this area is encouraged by widespread

applications of ASR, such as assistive tools for physically disabled people, railway stations, and customer service through voice-based interfaces. Conventional ASR systems may not be optimal for low-resource languages. In India, ASR is in its early stages of development in the aspect of recognition of utterances spoken by the various users in different languages of their region. Only the Hindi language has commercially accessible ASR devices till now and few Indian languages, including Hindi, Marathi, Malayalam, Tamil, Telugu, and Bengali, have well-developed ASRs, while others are still under development. Indian language voice recognition research has not yet reached that level of perfection. It's challenging to create a speech corpus for an Indian language. A significant challenge persists in low-resource languages, particularly those with limited availability of both audio and text data. This complexity is heightened by the existence of multiple dialects, as observed in Indian languages. Nonetheless, many Indian languages can be categorized into the same language families, sharing common scripts and grammatical structures. In such cases, adaptation and fine-tuning techniques become crucial for overcoming the inherent low-resource nature of the data by leveraging well-resourced languages that exhibit similarities. The speech research community has focused a lot of attention on ASR for under-resourced languages over the last decade.

One of the under-resource languages is the Punjabi Language, a very popular language in northern states of India, specifically Punjab, still the resources are insufficient. Creating an ASR system for children's speech in the Punjabi language poses additional challenges due to the absence of a dedicated children's speech corpus. Researchers tackling this linguistic domain find themselves compelled to undertake the laborious task of collecting their own corpus. This process is inherently demanding. Moreover, the scarcity of children's speech data imposes constraints on the adoption of state-of-the-art architectures like End-to-End (E2E) systems. The development of any ASR system using the E2E architecture necessitates a substantial volume of data, further complicating the task in the context of limited available corpus for children in Punjab. Table 9 represents the literature reported on the Punjabi ASR developed in the case of children. The summary of the acoustic mismatch approaches used for children's ASR is given in Table 10.

The summary shows the trends in the corpus, number of speakers, age groups, feature extraction, and acoustic modelling used by various research groups. The majority of the child corpus is in the age group of 3 to 14 years children. Further, the research data for preschool children is lacking in the literature, since collecting preschool children's data is very challenging and takes a considerable amount of time and effort. Moreover, as part of the data collection, people must go to toddlers' homes or places. Additionally, engaging these kids for recording their voice could be even more difficult task as they can't speak for a long time. However, Preschool children's data collection is still very demanding. Therefore, it is expected that researchers will be able to conduct research on preschool children in the future. The majority of the collected data that is publicly available is in English. Limited corpus of other languages is available to the public. As far as regional languages are concerned, limited children corpus is available, and for endangered languages, there is almost no child corpus. When it comes to feature extraction, MFCC is still the dominant technique in the field. The researchers also worked on other fused and hybrid techniques for feature extraction in their studies. Since features are not explicitly calculated in E2E architectures, so the focus of researchers is now shifting to E2E systems. A unified system is created to handle all steps involved in the E2E architecture to streamline the process. Though, some research groups started opting for E2E systems, however, still hybrid architecture, e.g. HMM-DNN, are widely used by many researchers.

Table 9 Survey of children’s ASR systems in the Punjabi language

Reference	Children/ Age Group (in years)	Feature Extrac- tion Technique	Acoustic Model- ling	Normalization (if used)	Adaptation Model	Data Augmentation Techniques	Accuracy/ WER	Remarks
[28]	32 Speakers	MFCC, GFCC, Pitch	DNN-HMM	VTLN CMVN	MLLT	three different noise were used: factory, babble, white	relative improvement of 30.94%	Pitch features were extracted from Punjabi data set, and the test set was normalized using a VTLN approach to accom- modate the tonal characteris- tics of the language
[178]	31Speakers 7–13	MFCC	DNN-HMM	VTLN, Pitch and Duration	fMLLR	prosody-modifica- tion-based data augmentation	RI of 32.1%	Error rate reduced by applying prosody-modification-based out-of-domain data augmenta- tion
[176]	31 Speaker	MFCC	Discriminative training with MMI, bMMI, fMMI, and fbMMI varia- tions	—	—	—	22–26% of RI	-The discriminative techniques of bMMI have been tested in context of dependent and independent models -Variety of parameters has been combined to optimize bMMI and fbMMI acoustic models
[173]	39 Speaker	MFCC	DNN-HMM, SGMM	—	MLLT, SAT, fMLLR	—	improvement of 0.03% to 16.47% WER	Improvement in Children speech recognition system obtained using pitch enhancement features under disparate conditions
[177]	66 Speaker 5–13	Frequency Domain Linear Predic- tion (FDLP), MFCC MFCC-FDLP hybrid approach	Time Delay Neural Network (TDNN)	VTLN	SRA	Tacotron 2, an end-to-end Text to Speech (TTS) generative model has been used	relative improvement of 29.44%	-Tacotron 2 offers an innovative approach to improve ASR for children with low resource languages -Speaker Adaptive Training has been done for unseen speakers to increase the robustness of the model

Table 9 (continued)

Reference	Children/ Age Group (in years)	Feature Extrac- tion Technique	Acoustic Model- ling	Normalization (if used)	Adaptation Model	Data Augmentation Techniques	Accuracy/ WER	Remarks
[145]	—	MFCC, GFCC and hybrid MF-GFCC	GMM-HMM DNN-HMM	VTLN, Pitch	—	in-domain data augmentation with synthetic datasets using factory, speech babble and white noise	relative improve- ment (RI) of 20.59% on noisy and 19.39% on clean environ- ment conditions	Fused feature extraction approach to handle large training complexity using MF-GFCC technique, through feature warping method, has been explored
[172]	39 speakers 8–14 years	MFCC	DNN DNN + HMM	Pitch	—	using Tacotron	RI of 9–12%	MFCC + pitch feature extraction and DNN acoustic model- ling was used on Tacotron synthesized speech data, merged with the available speech corpora, to improve Punjabi ASR
[171]	38 speakers 8–15 yrs	MFCC	Subspace Gauss- ian mixture model (SGMM) SGMM-HMM	—	—	—	WER 16.34%	-SGMM acoustic modelling technique is used to develop Punjabi children ASR -WER reduced remarkably using SGMM by varying the feature dimensions
[180]	39 Seapkers 8 to 13	MFCC	DNN- HMM	—	—	—	RI of 14%	There was a significant improve- ment in tonal and non-tonal word recognition in Punjabi speech for children when the POV and f0 features were combined

Table 9 (continued)

Reference	Children/ Age Group (in years)	Feature Extrac- tion Technique	Acoustic Model- ling	Normalization (if used)	Adaptation Model	Data Augmentation Techniques	Accuracy/ WER	Remarks
[179]	30 speakers 6–17 years	MFCC	GMM-HMM DNN-HMM	VTLN, LP-PSOLA Pitch and Duration	MLLR, fMLLR	—	WER is reduced by 21.18%	-Pitch, and duration prosody modifications are used to reduce acoustic mismatches -The prosody-based ASR system provided better recognition rates compared to the baseline system
[29]	7 to 13	MFCC + pros- ody features	HMM with discriminative techniques like MMI, BMIMI, FMIMI	—	—	out-domain data augmentation	RI of 20–25% was observed	-Out domain data augmentation techniques used to enhance training speech data -Accuracy children Punjabi- ASR was improved to 88% when integrated prosodic features were extracted from data augmented corpora

Table 10 Summary of acoustic mismatch approaches used for children’s ASR

Ref	Language/ Corpus/Data set	No. of Children/ Age Group (in years)	Feature Extraction Technique	Acoustic Modeling
[69]	TIDIGITS Corpus	51 Speakers	PLP	LPC ceptral analysis
[50]	Lang: Danish Rafael.0 telephone speech database	306 Speakers/ 8 to 18	—	HMM
[73]	Two task-independent speech databases (con- sisted of Italian read sentences)	7 to 13 years	MFCC	HMM
[74]	Lang: Swedish PF-STAR	198 speakers/ 4 to 8	MFCC	HMM-GMM
[75]	TIDIGITS Corpus	10 Speakers/ 6 to 15	MFCC	HMM-GMM
[76]	ChildIt and HUB4	7 to 13	MFCC, PLP, RPLP	HMM-GMM
[84]	Lang: English, TIDIGITS, Isolated digits and digit strings	51 speakers/ 7 to14	MFCC	HMM
[47]	Three corpora were used: 1. Lang: Italian ChildIt 2. Italian SpontIt 3. American English CID	1.171 Speakers/ 7 to 13 2. 21 Speakers/ 8 to 12 3.436 Speakers/ 5 to 18	MFCC	HMM
[85]	Lang: English, Italian Collected data was kept in two databases 1.Native English 2.Non-native Italian	1.7 to12 2.10 to 11	MFCC	HMM
[77]	Read data collected	Grade 3 to grade 5	MFCC, PMVDR	HMM
[86]	Lang: British English PFSTAR	122 Speaker/ 3 to 14	PLPCC,SMAC,PN-SMAC	DNN,SGMM
[87]	OGI Kids	100 Speakers/ Kindergarten to 10th	MFCC	DNN-HMM

Table 10 (continued)

Ref	Language/ Corpus/Data set	No. of Children/ Age Group (in years)	Feature Extraction Technique	Acoustic Modeling
[79]	Lang: British English PFSTAR	122 Speakers/ 4 to 13	MFCC	HMM-GMM DNN-HMM TDNN
[80]	Lang: British English PFSTAR	122 Speakers/ 4 to 13 years	MFCC, PLPCC	HMM-GMM DNN-HMM
[81]	PFSTAR	122 Speakers/ 4 to 13	MFCC	DNN-HMM
[82]	CMU kids Corpus	151 Speakers/ 6 to 11	MFCC	Filter based Discriminative autoen- coder (f-DeAE), TDNN
[83]	Lang: British English PFSTAR	60 Speakers/ 4 to 14	MFCC	DNN-HMM
[78]	OGI Kids` Speech Corpus	151 Speakers/ Kindergarten to 10th	MFCC	BLSTM-based
[49]	Lang: German Corpus 3.5 h	62 Speakers/ 10 to 12	MFCC	—
[71]	1.Lang: native: German & Swedish 2.Long:non-native: English PFSTAR	150 Speakers/ 4 to 15	—	HMM
[92]	OGI Corpus	6 to 18 years	MFCC	HMM
[93]	Lang: Punjabi Own collected data se	39 Speakers	MFCC, LPCC, RASTA-PLP,GFCC	DNN-HMM
[94]	JWC ehild database	24 Speakers/ 10 to 11	(CS-ACELP)	DTW
[95]	PFSTAR	150 Speakers/ 4 to 15	PACT-MFCC, NUSS-MFCC	HMM-GMM, LSTM
[74]	PFSTAR	198 Speakers/ 4 to 8	MFCC	GMM-HMM

Table 10 (continued)

Ref	Language/ Corpus/Data set	No. of Children/ Age Group (in years)	Feature Extraction Technique	Acoustic Modeling
[108]	Lang: British English PFSTAR	122 Speakers/ 3 to 14 years	MFCC	Context-dependent HMM, DNN-HMM
[102]	Lang: British English PFSTAR	122 Speakers/ 3 to 14 years	MFCC	GMM-HMM
[97]	Lang: British English PFSTAR	122 Speakers/ 3 to 14 years	MFCC	GMM-HMM
[110]	Lang: British English PFSTAR	122 Speakers/ 3 to 14 years	MFCC	GMM-HMM, DNN-HMM
[111]	Lang: British English PFSTAR	122 Speakers/ 3 to 14 years	MFCC	GMM-HMM, DNN-HMM
[104]	Lang: British English PFSTAR	122 Speakers/ 3 to 14 years	E2E	Transfer Learning
[106]	Lang: British English PFSTAR	158 Speakers/ 4 to 14	CNN-based E2E System	CNN-based E2E System

9 Conclusion

A comprehensive review of the literature on children's ASR has been presented in this article. Based on the reviewed literature, it has been observed that a lot of work on children's ASR in the field of reading tutors, foreign language learning tools, computer games, education, etc. has been carried out by various research groups; however, limited work has been reported in low resource languages. The improvements reported in children ASR are still not up to the level it has been reported for adult-ASR. The inaccessibility of child corpora publicly is significant barrier to children's ASR. Hybrid feature extraction techniques have not been explored much in the case of children's ASR. Acoustic and linguistic variabilities contribute to the degradation of any child's ASR system. Enough work has been reported on the acoustic variabilities of children; however, linguistic variabilities have not been explored much. There is a scope for improvement in continuous and spontaneous children's speech since not much has been reported in the existing literature. Further, the work reported on infants' speech is also in its early stages of development.

Variability of children speakers, non-availability of child corpus, and difficulties in child speech data collection are some of the major challenges responsible for less children-ASR. These challenges become more acute in the case of low-resource languages, particularly Punjabi. The work reported specifically in the Punjabi language is in the infant stage. Most ASR systems reported in the Punjabi Language have been focused on adult speech; however, limited work has been reported on children's speech corpus. The major reason for the non-availability of Punjabi-ASR is the lack of availability of Punjabi children's speech corpus publicly. A significant obstacle lies in the challenge of finding native speakers who possess the required technical skills for developing ASR systems in their native languages. Further, data augmentation techniques with pooled datasets have not been explored much in child ASR systems in the Punjabi language. Speaker variabilities such as age, speaking rate, gender etc., are some of the essential parameters that need to be explored to better understand the Punjabi ASR system.

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Declarations

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